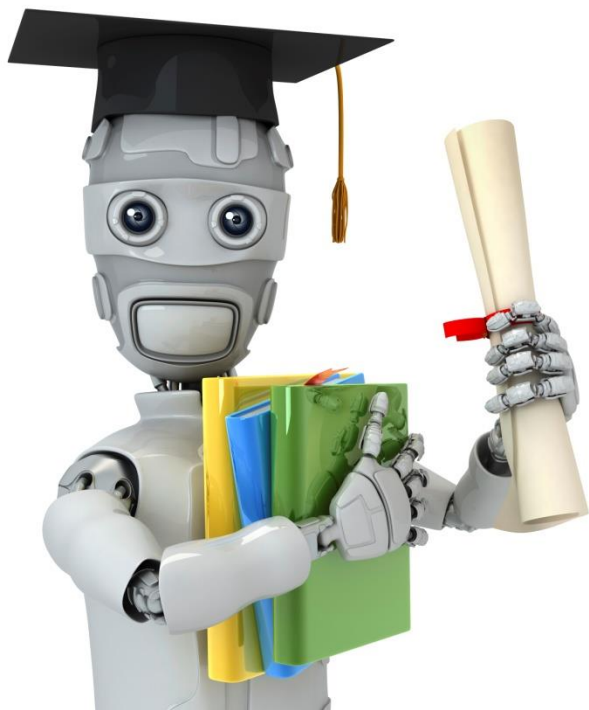




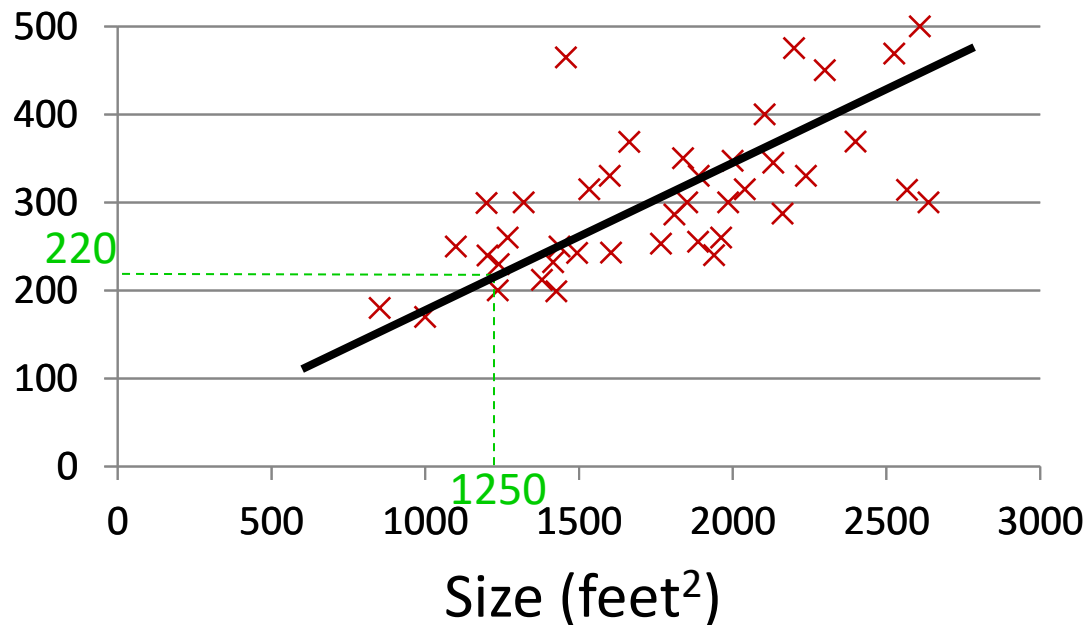
Machine Learning

Linear regression with one variable

Model representation

Housing Prices (Portland, OR)

Price
(in 1000s
of dollars)



Supervised Learning

Given the “right answer” for each example in the data.

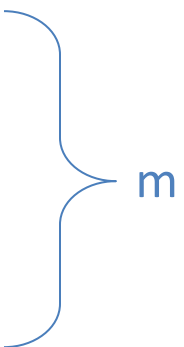
Regression Problem

Predict real-valued output

Classification : Discrete-valued output

Training set of housing prices (Portland, OR)

Size in feet ² (x)	Price (\$) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...



Notation:

m = Number of training examples

x 's = "input" variable / features

y 's = "output" variable / "target" variable

(x, y) – one training example

$(x^{(i)}, y^{(i)})$ – i^{th} training example

$$x^{(1)} = 2104$$

$$x^{(2)} = 1416$$

$$y^{(1)} = 460$$

Training Set



Learning Algorithm



Size of
house
 x



h

hypothesis

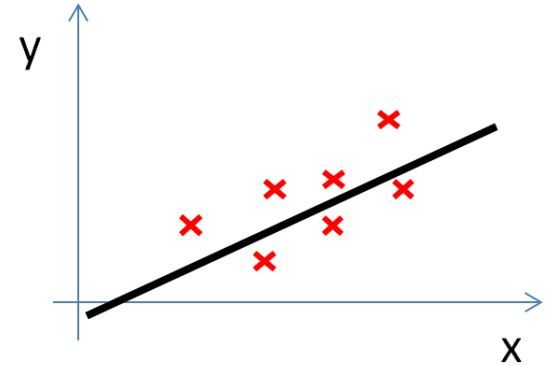


Estimated
price
Estimated
value

h maps from x 's to y 's

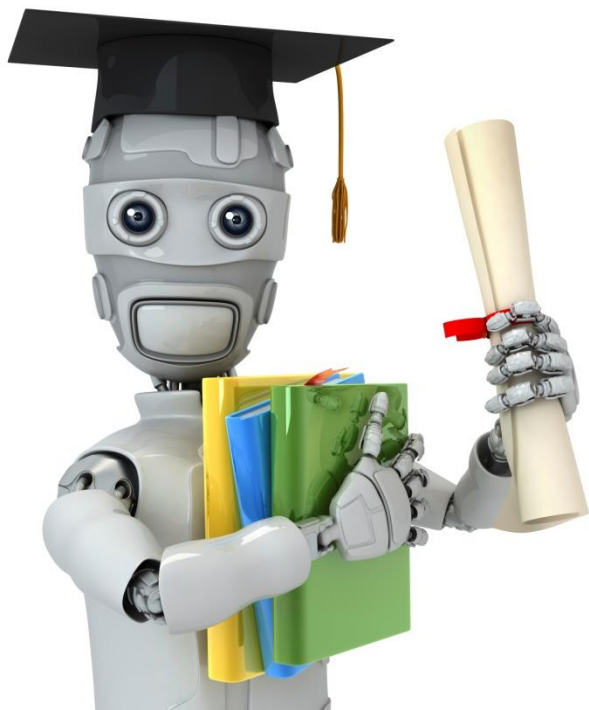
How do we represent h ?

$$h_{\theta}(x) = \Theta_0 + \Theta_1 x$$



Linear regression with one variable.
Univariate linear regression.

One variable



Machine Learning

Linear regression with one variable

Cost function

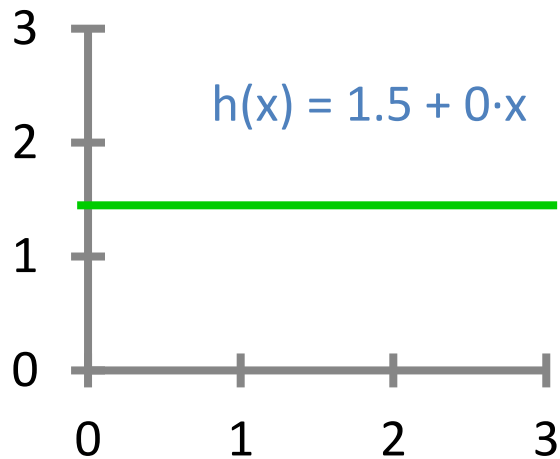
Training Set	Size in feet ² (x)	Price (\$) in 1000's (y)
	2104	460
	1416	232
	1534	315
	852	178
	...	...

Hypothesis: $h_{\theta}(x) = \theta_0 + \theta_1 x$

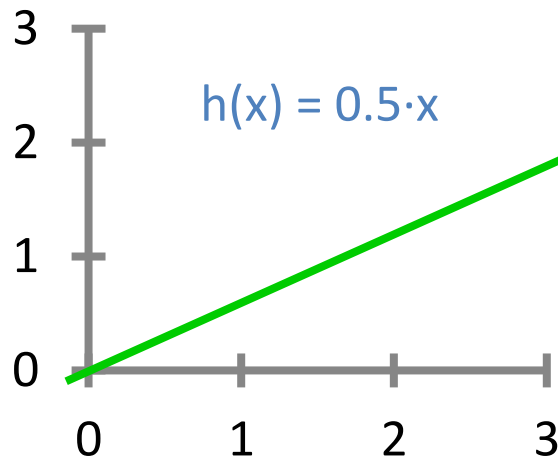
θ_i 's: Parameters

How to choose θ_i 's ?

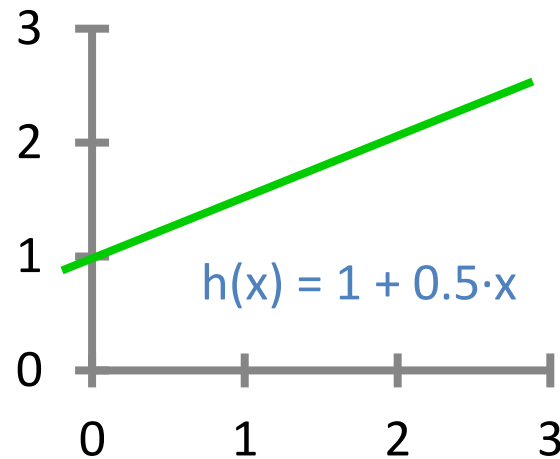
$$h_{\theta}(x) = \theta_0 + \theta_1 x$$



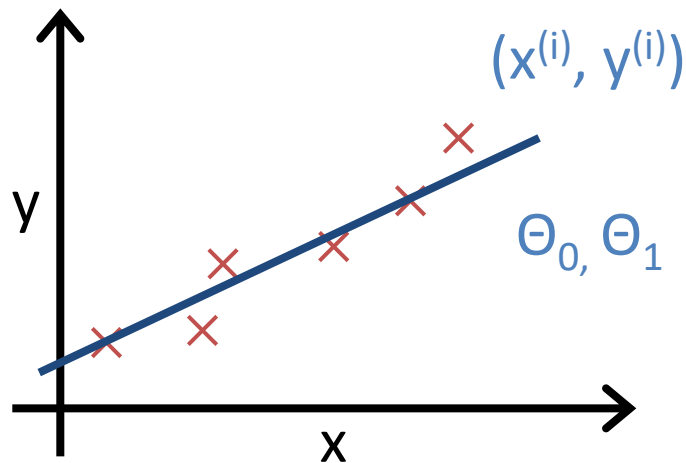
$$\theta_0 = 1.5$$
$$\theta_1 = 0$$



$$\theta_0 = 0$$
$$\theta_1 = 0.5$$



$$\theta_0 = 1$$
$$\theta_1 = 0.5$$



Idea: Choose θ_0, θ_1 so that $h_{\theta}(x)$ is close to y for our training examples (x, y)

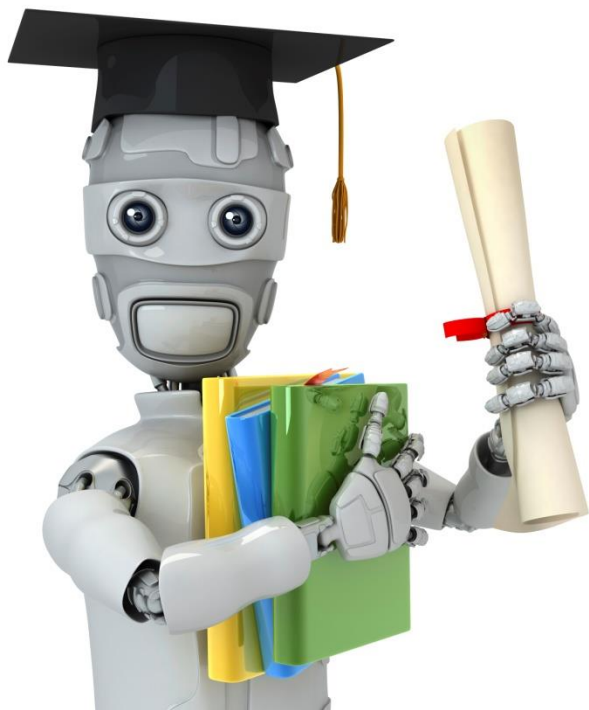
$$\underset{\theta_0, \theta_1}{\text{minimize}} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$h(x) = \theta_0 + \theta_1 x^{(i)}$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Minimize $J(\theta_0, \theta_1)$: Cost Function
 θ_0, θ_1

Squared error function



Machine Learning

Linear regression
with one variable

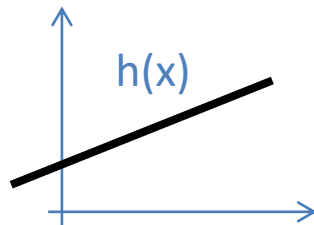
Cost function
intuition I

Hypothesis:

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

Parameters:

$$\theta_0, \theta_1$$



Cost Function:

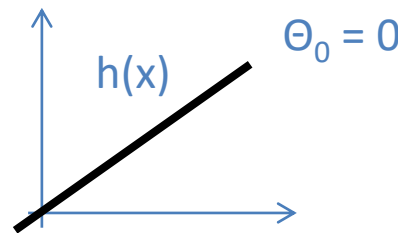
$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Goal: minimize $J(\theta_0, \theta_1)$
 θ_0, θ_1

Simplified

$$h_{\theta}(x) = \theta_1 x$$

$$\theta_1$$

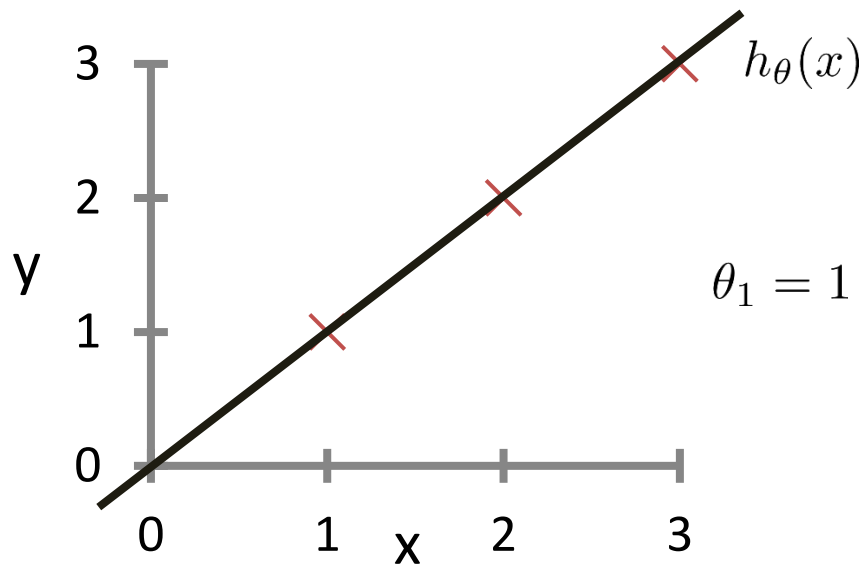


$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

minimize $J(\theta_1)$
 θ_1

$$h_{\theta}(x)$$

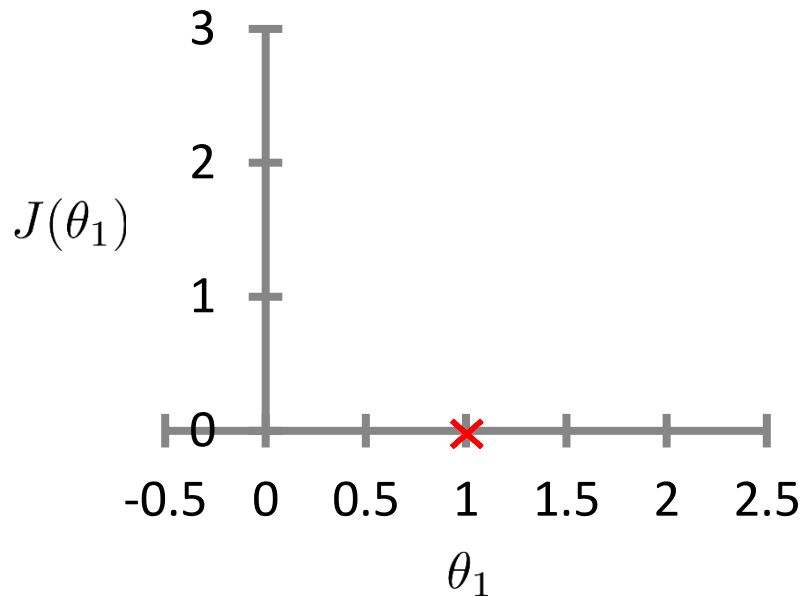
(for fixed θ_1 , this is a function of x)



$$\begin{aligned} J(\Theta_1) &= \frac{1}{2m} \sum_{i=1}^m (h_{\Theta}(x^{(i)}) - y^{(i)})^2 \\ &= \frac{1}{2m} \sum_{i=1}^m (\Theta_1 x^{(i)} - y^{(i)})^2 \\ &= \frac{1}{2m} (0^2 + 0^2 + 0^2) \end{aligned}$$

$$J(\theta_1)$$

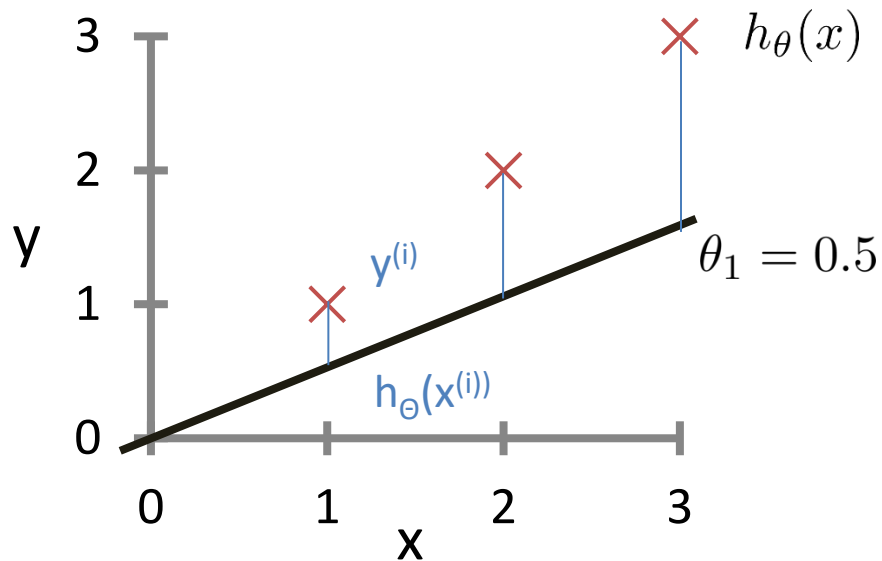
(function of the parameter θ_1)



$$J(1) = 0$$

$$h_{\theta}(x)$$

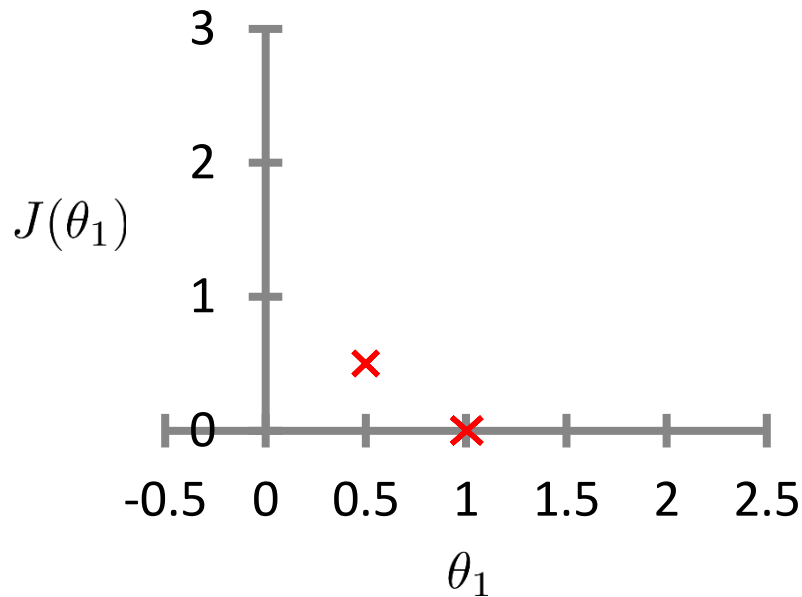
(for fixed θ_1 , this is a function of x)



$$\begin{aligned} J(0.5) &= \frac{1}{2 \cdot 3} \sum_{i=1}^3 [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2] \\ &= \frac{1}{6} \cdot (3.5) = 0.58 \end{aligned}$$

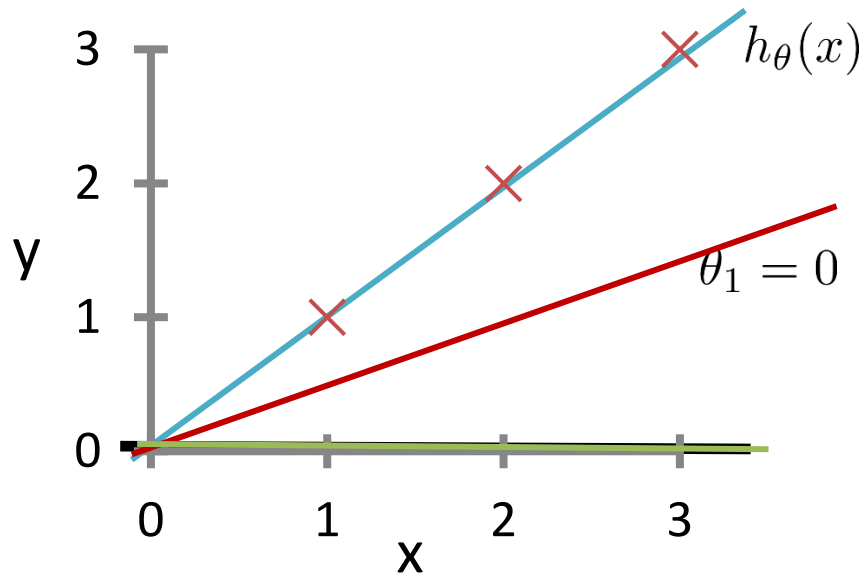
$$J(\theta_1)$$

(function of the parameter θ_1)



$$h_{\theta}(x)$$

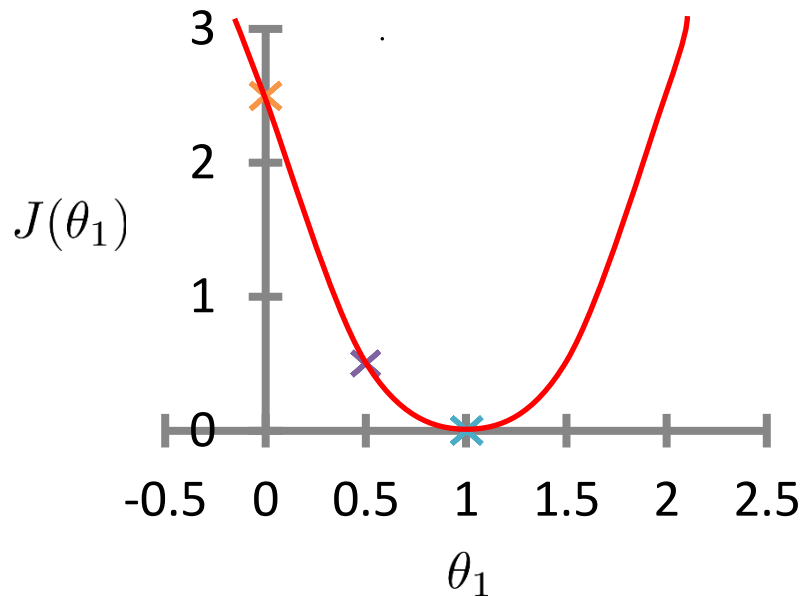
(for fixed θ_1 , this is a function of x)

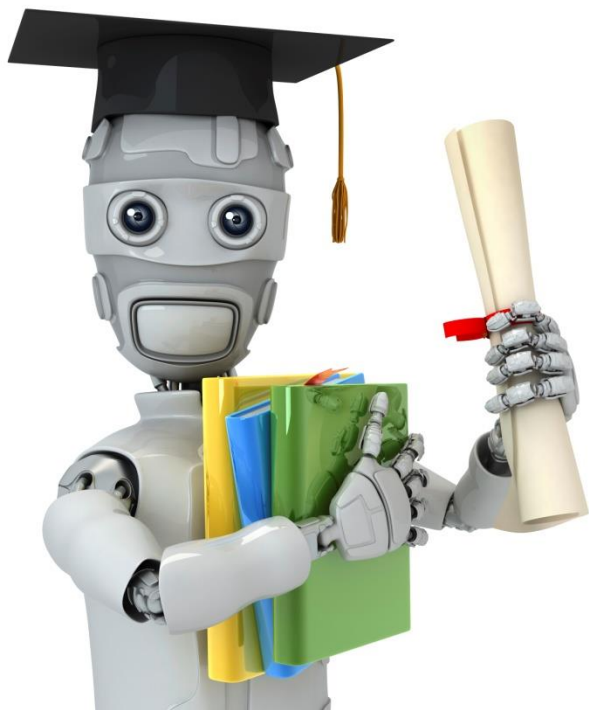


$$\begin{aligned} J(0) &= \frac{1}{2 \cdot 3} \sum_{i=1}^3 [1^2 + 2^2 + 3^2] \\ &= \frac{1}{6} \cdot 14 = 2.3 \end{aligned}$$

$$J(\theta_1)$$

(function of the parameter θ_1)





Machine Learning

Linear regression
with one variable

Cost function intuition II

Hypothesis: $h_{\theta}(x) = \theta_0 + \theta_1 x$

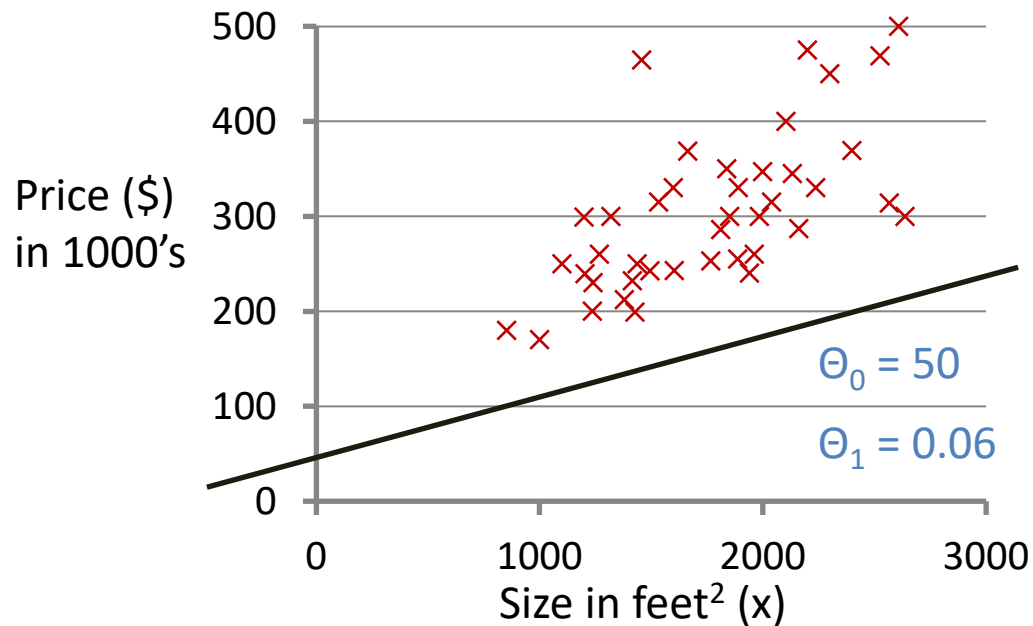
Parameters: θ_0, θ_1

Cost Function: $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$

Goal: minimize $J(\theta_0, \theta_1)$
 θ_0, θ_1

$$h_{\theta}(x)$$

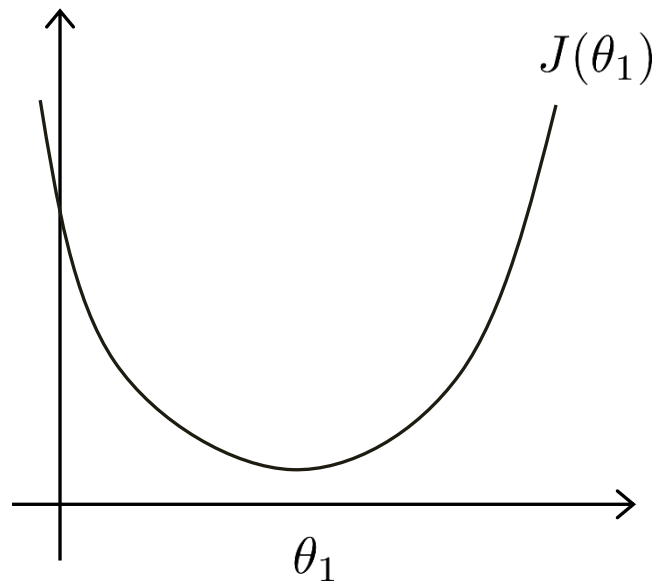
(for fixed θ_0, θ_1 , this is a function of x)



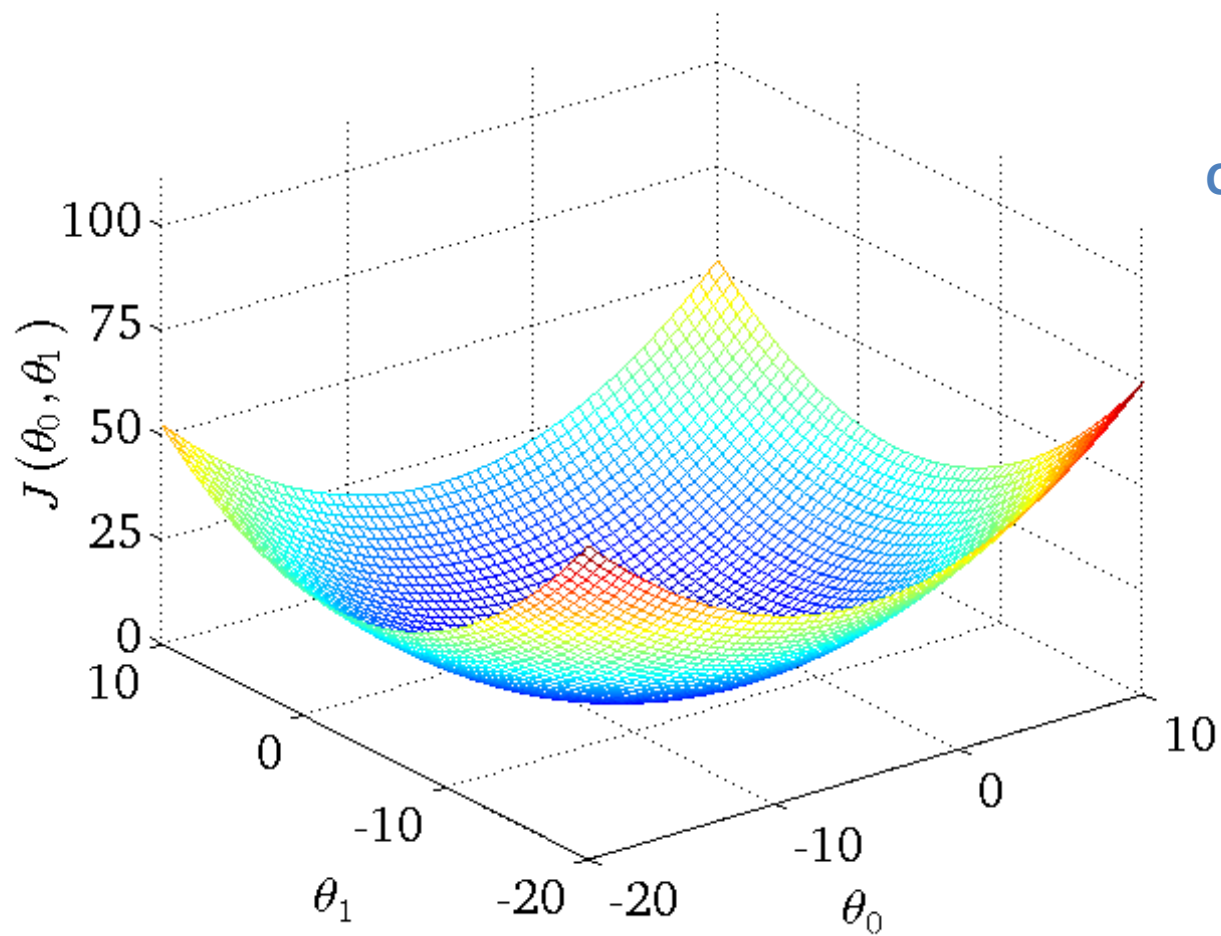
$$h_{\theta}(x) = 50 + 0.06x$$

$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)

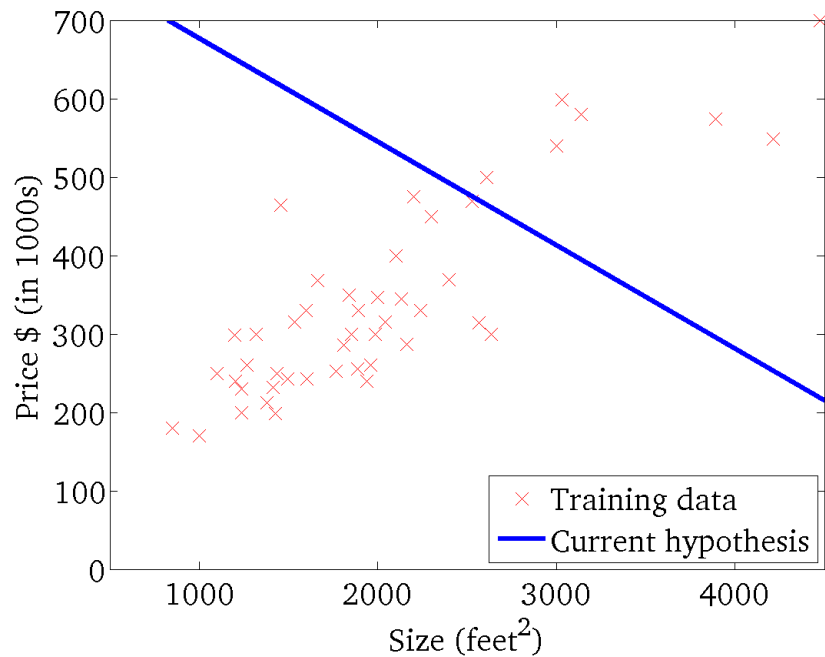


Contour plots



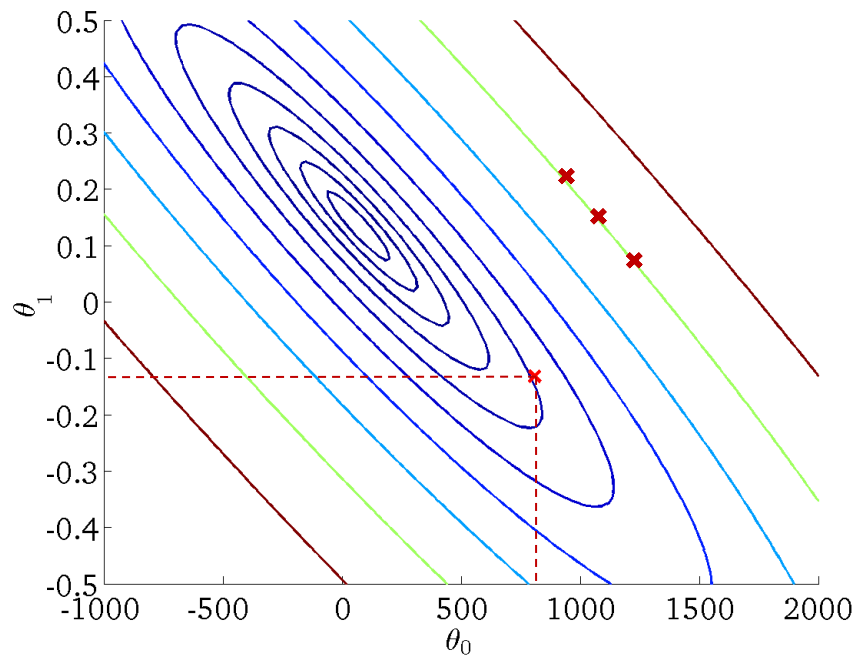
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



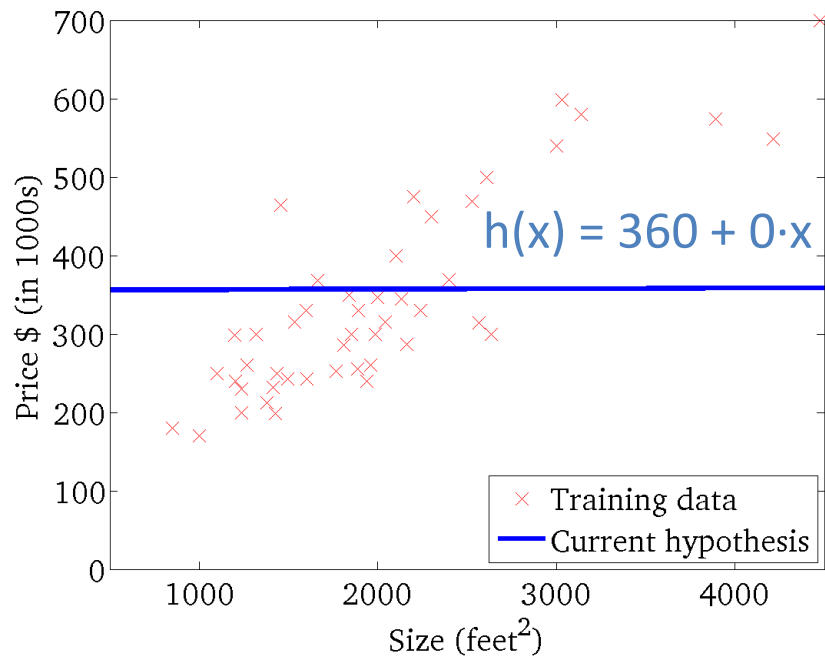
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



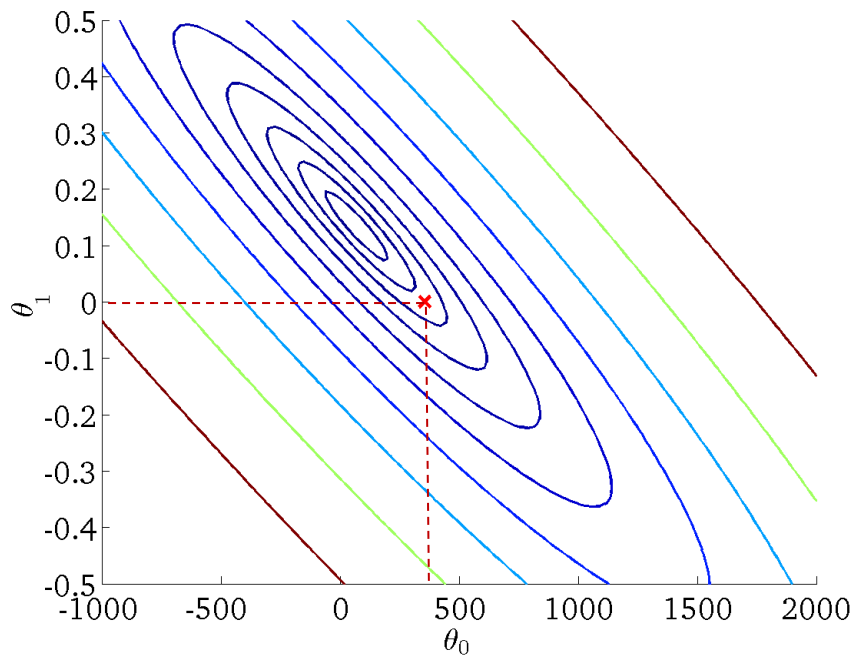
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)

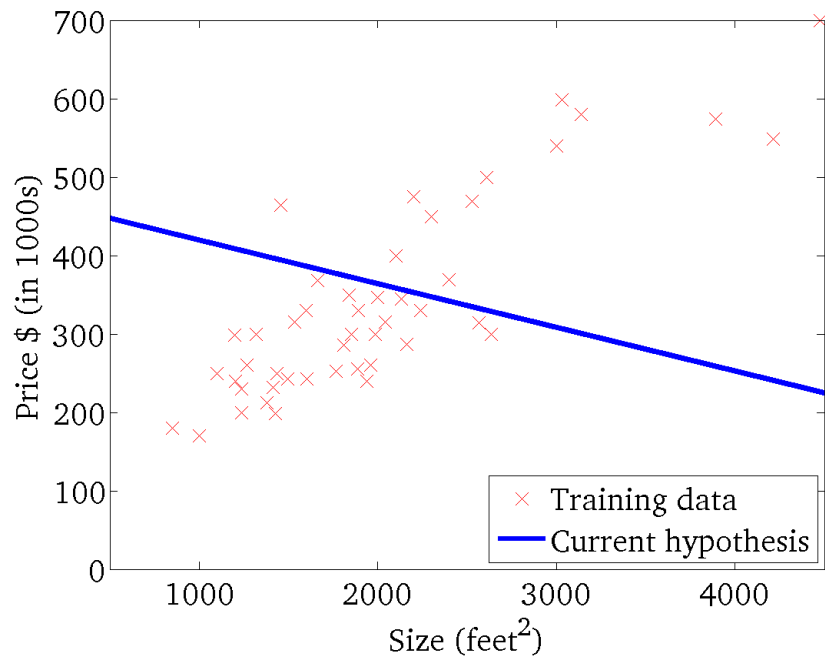


$$\Theta_0 = 360$$

$$\Theta_1 = 0$$

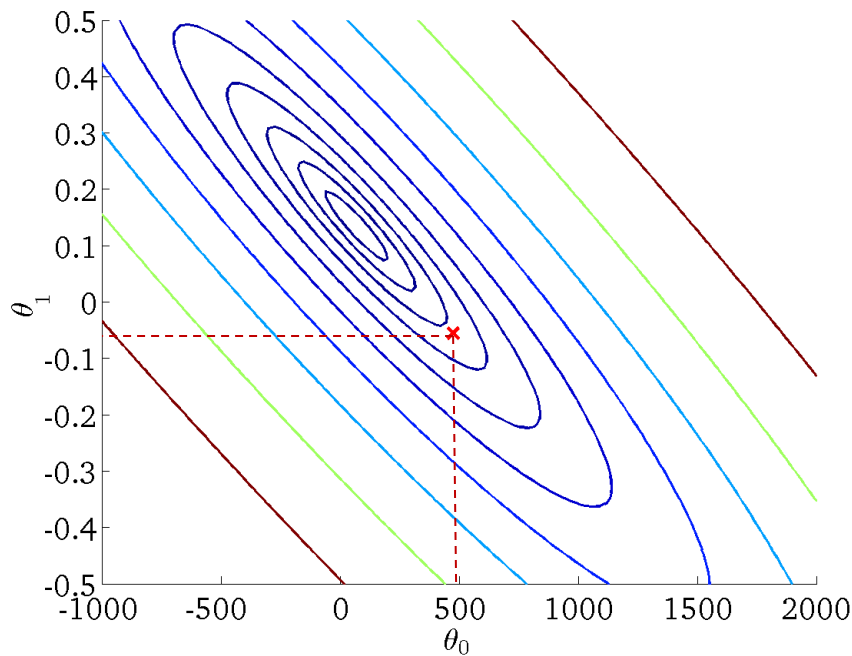
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



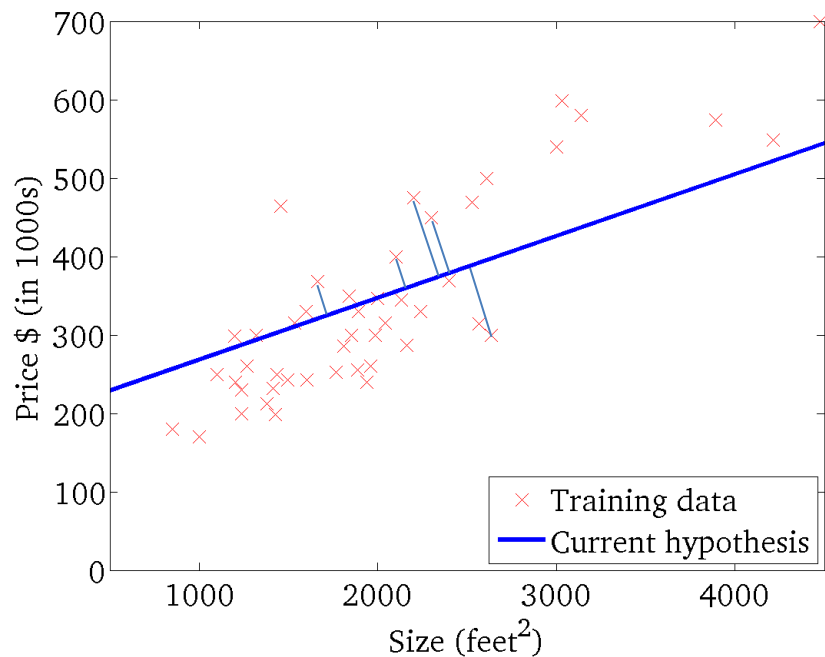
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



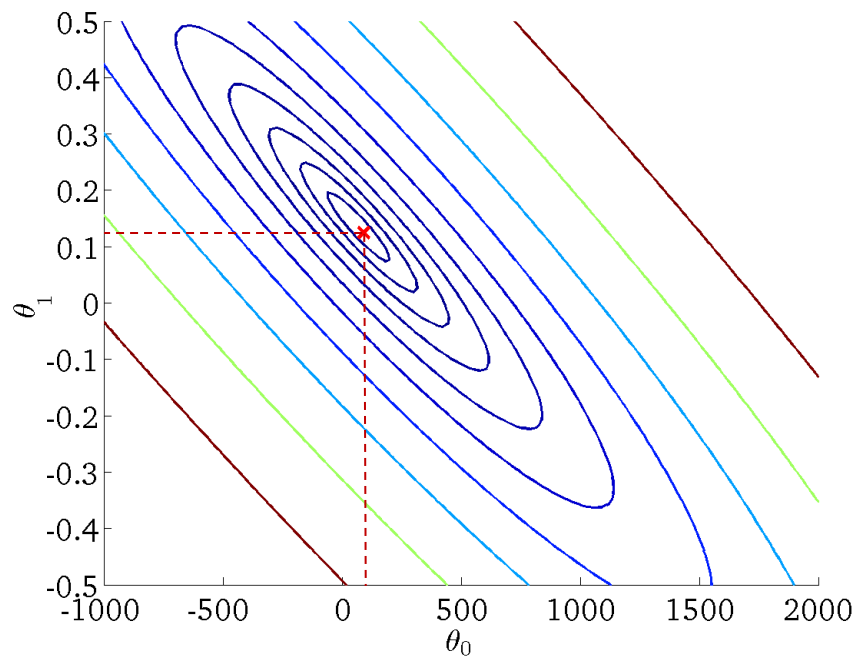
$$h_{\theta}(x)$$

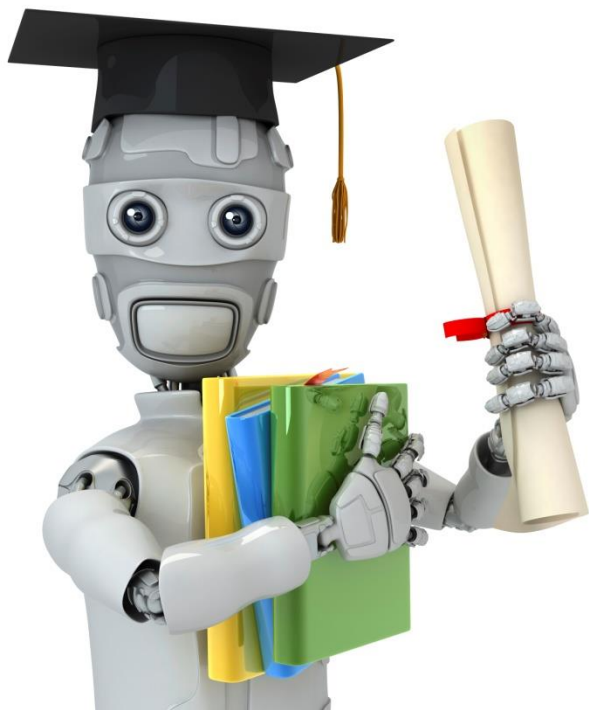
(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)





Machine Learning

Linear regression
with one variable

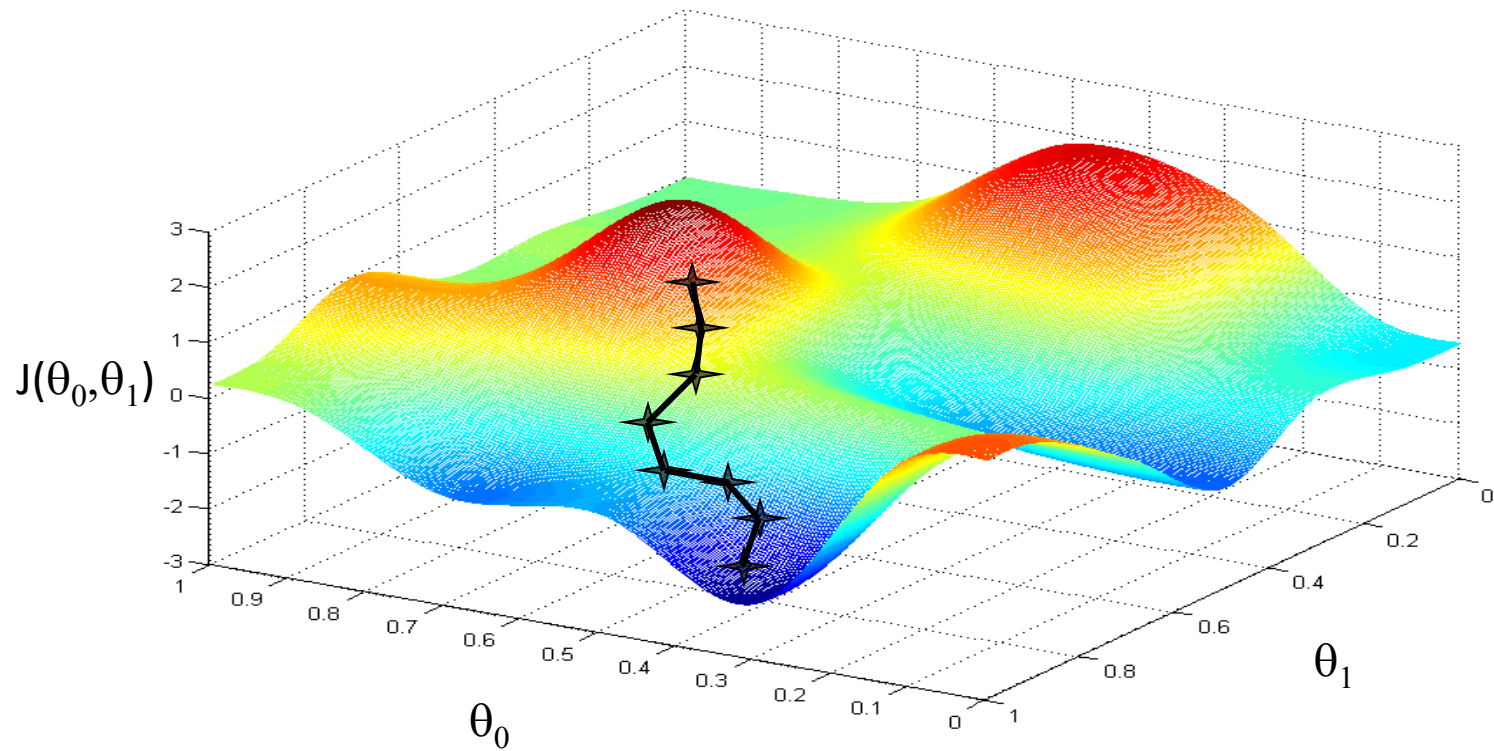
Gradient
descent

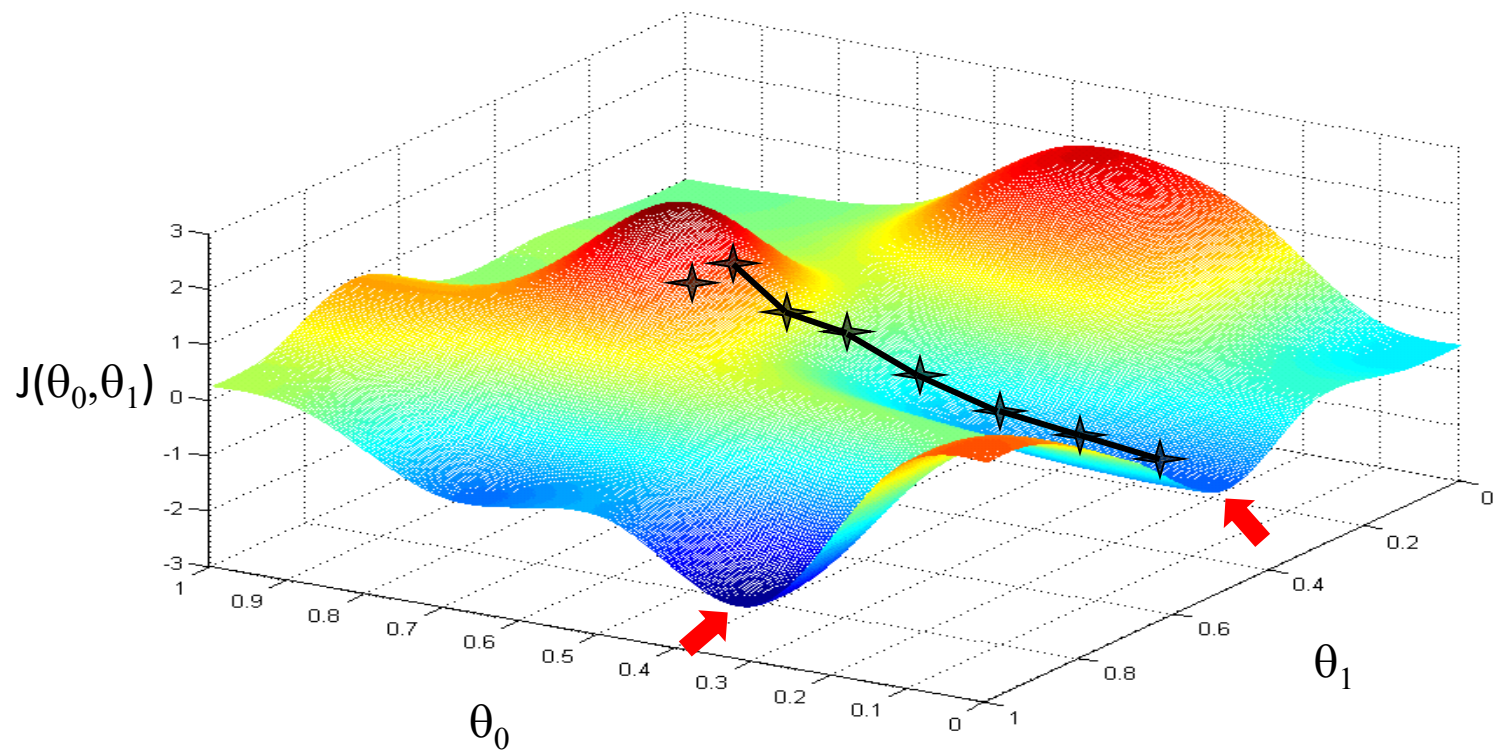
Have some function $J(\theta_0, \theta_1)$

Want $\min_{\theta_0, \theta_1} J(\theta_0, \theta_1)$

Outline:

- Start with some θ_0, θ_1
- Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$
until we hopefully end up at a minimum





Gradient descent algorithm

assignment

a:=b


repeat until convergence {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) \quad (\text{for } j = 0 \text{ and } j = 1)$$

}

Learning rate

Simultaneously update

θ_0 & θ_1

Correct: Simultaneous update

$$\text{temp0} := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\text{temp1} := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_0 := \text{temp0}$$

$$\theta_1 := \text{temp1}$$

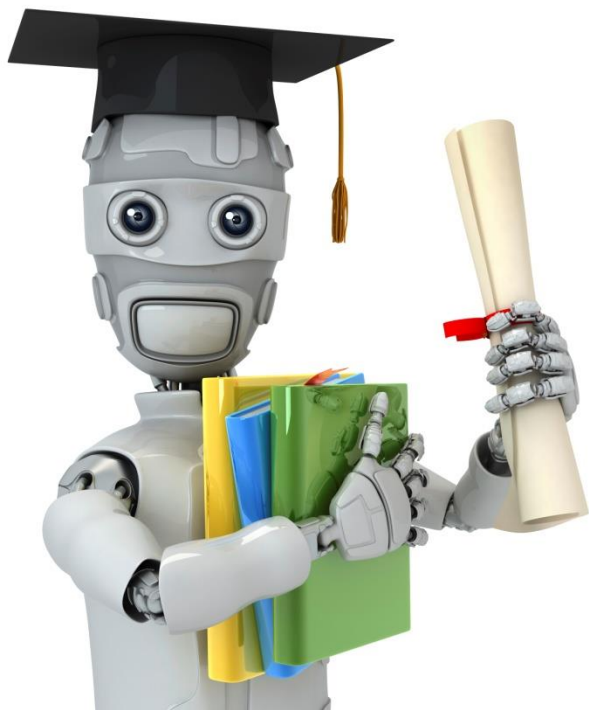
Incorrect:

$$\text{temp0} := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_0 := \text{temp0}$$

$$\text{temp1} := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_1 := \text{temp1}$$



Machine Learning

Linear regression with one variable

Gradient descent intuition

Gradient descent algorithm

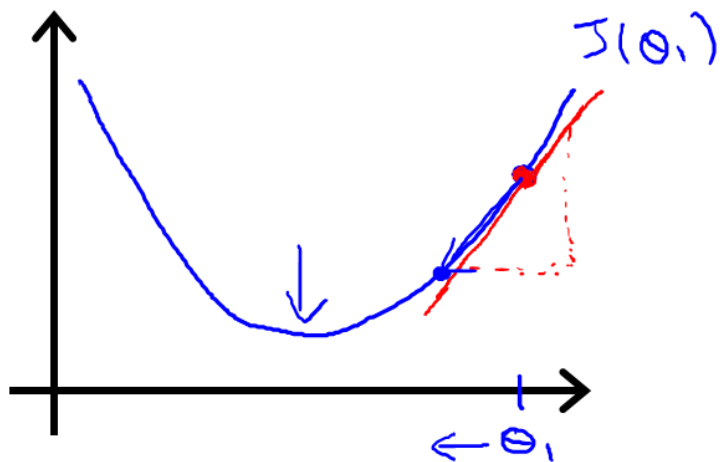
repeat until convergence {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

} Learning rate

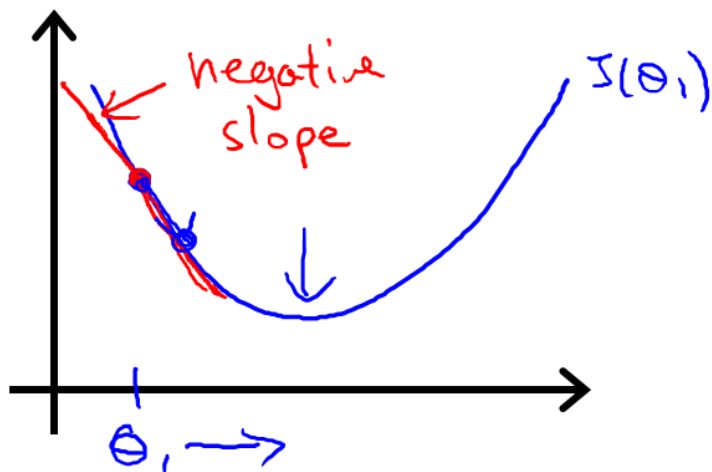
derivative

(simultaneously update
 $j = 0$ and $j = 1$)



$$\Theta_1 := \Theta_1 - \alpha \boxed{\frac{d}{d\theta_1} J(\theta_1)}_{\geq 0}$$

$$\Theta_1 := \Theta_1 - \alpha \cdot J(\text{positive number})$$



$$\Theta_1 := \Theta_1 - \alpha \boxed{\frac{d}{d\theta_1} J(\theta_1)}_{\leq 0}$$

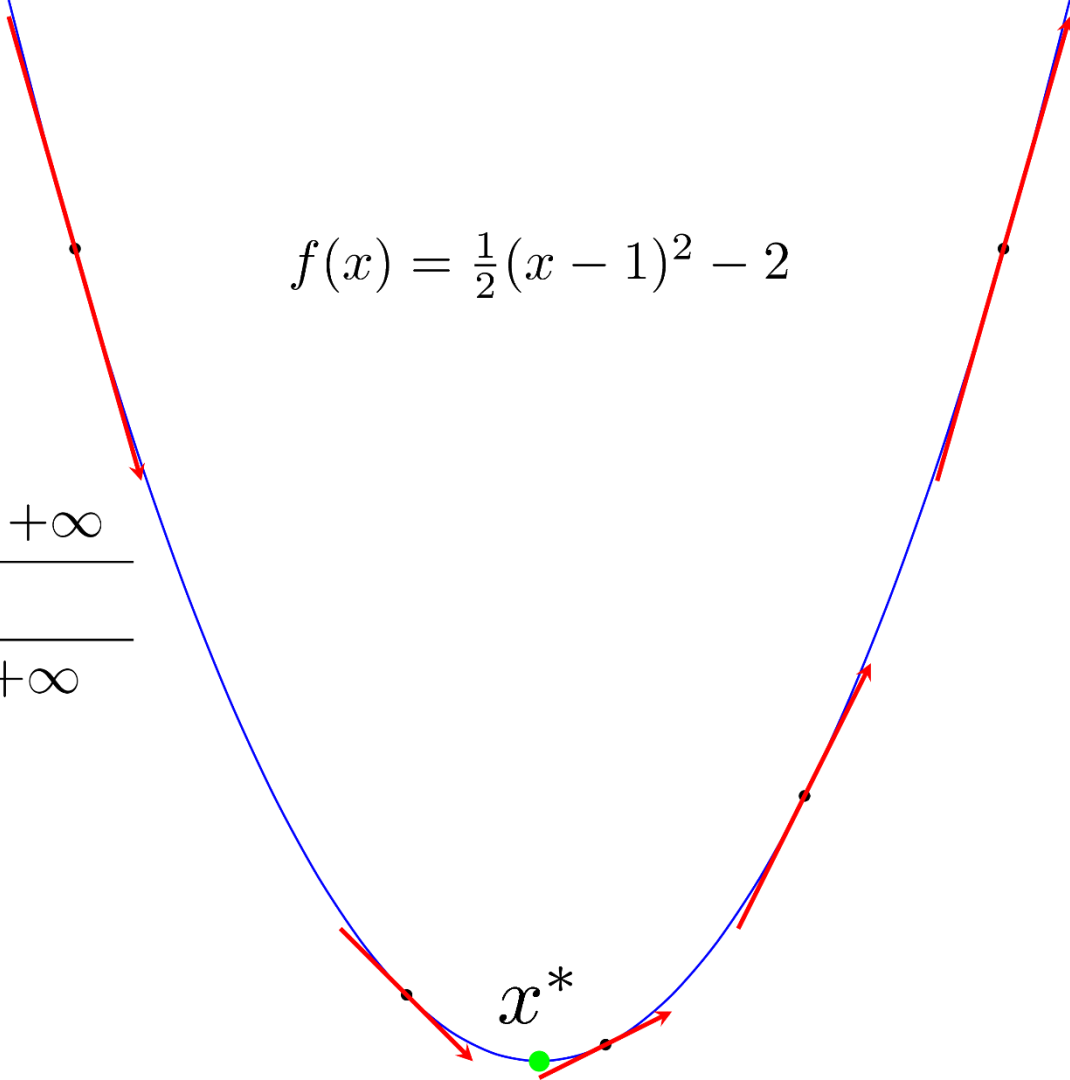
$$\Theta_1 := \Theta_1 - \alpha \cdot J(\text{negative number})$$

Figure credit: Vũ Hữu Tiệp

x	$-\infty$	1	$+\infty$
$f'(x)$	$-$	0	$+$
$f(x)$	$+\infty$		$+\infty$

$\swarrow$ $\swarrow$
 -2

$$f(x) = \frac{1}{2}(x - 1)^2 - 2$$



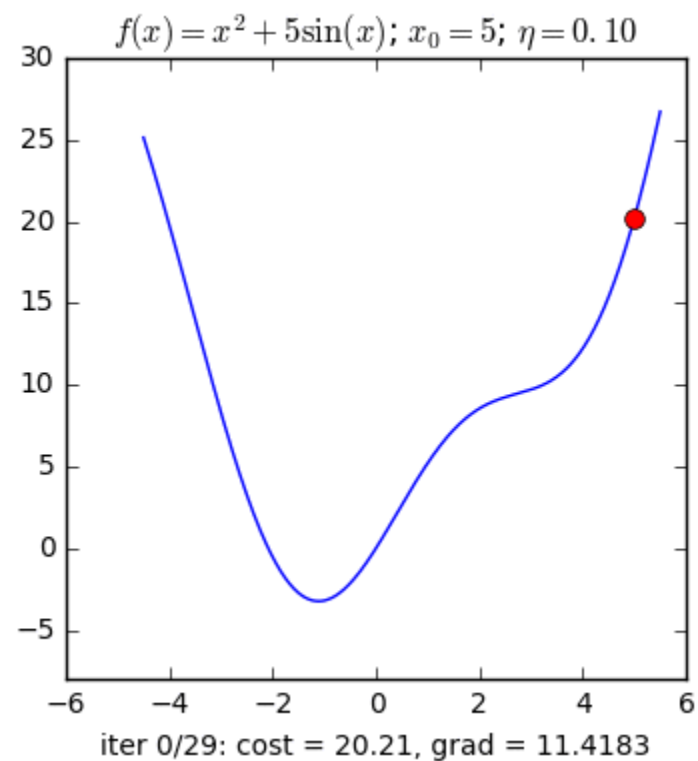
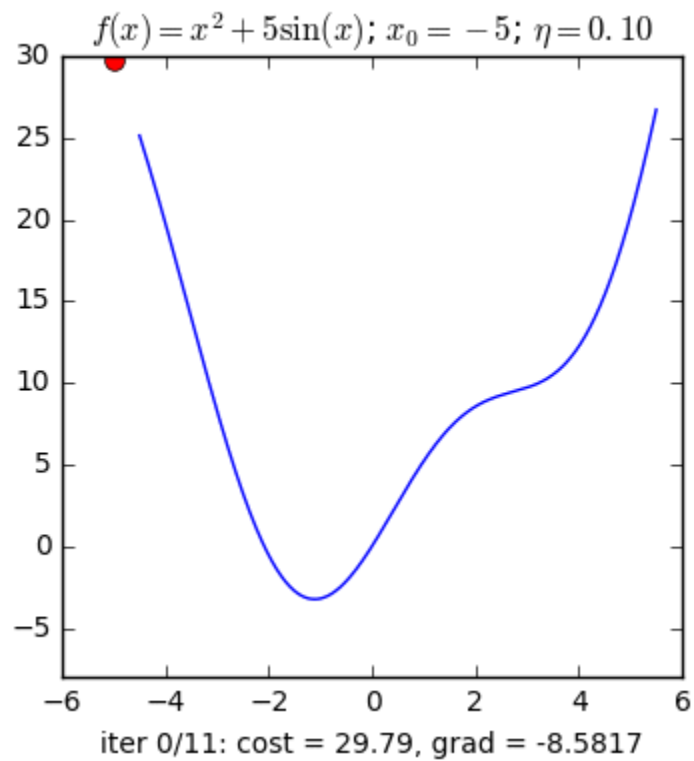
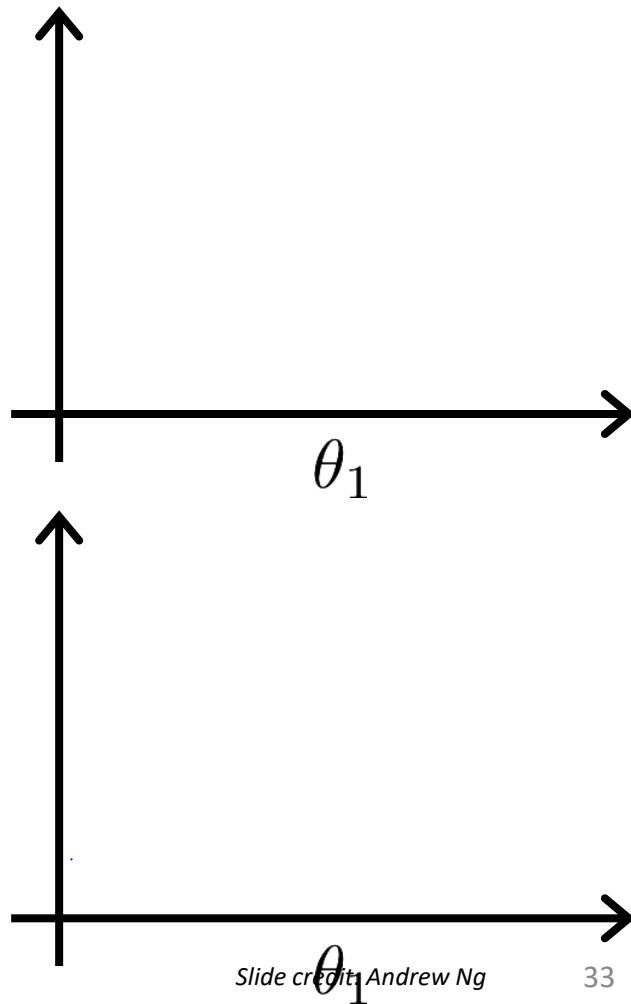


Figure credit: Vũ Hữu Tiệp

$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_1)$$

If α is too small, gradient descent can be slow.

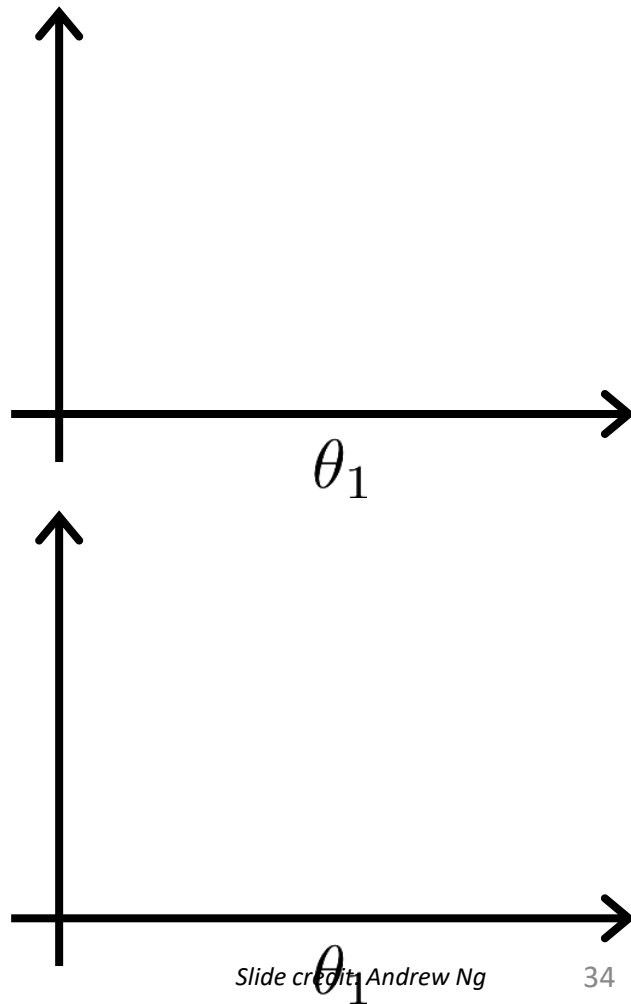
If α is too large, gradient descent can overshoot the minimum. It may fail to converge, or even diverge.

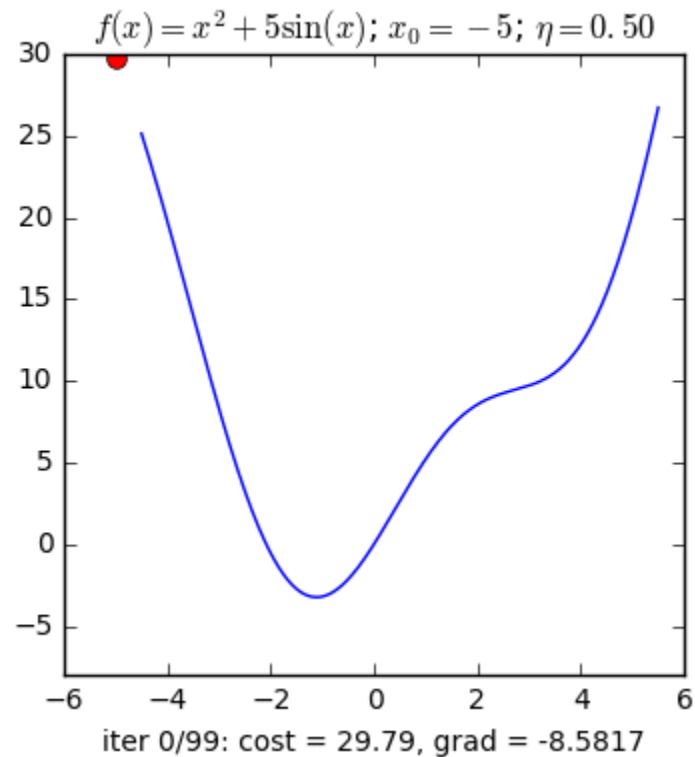
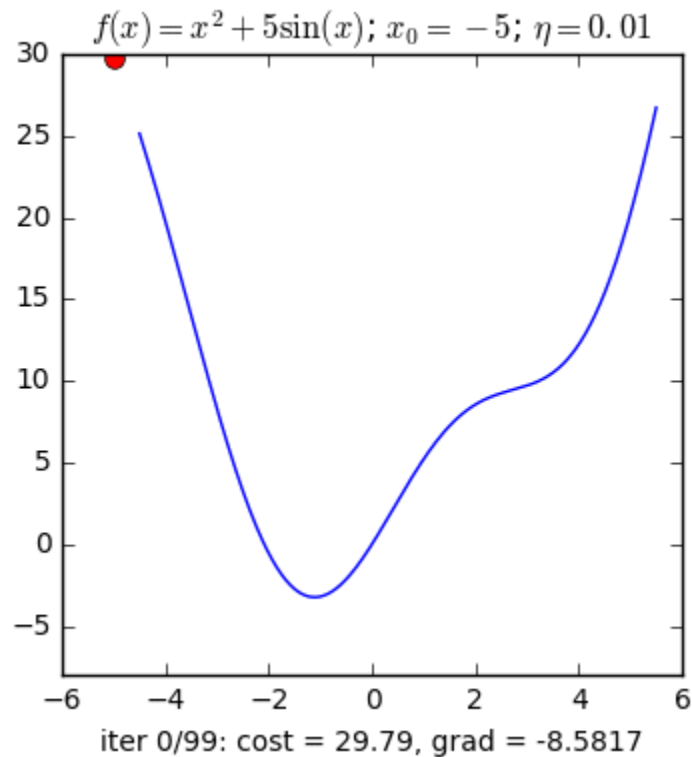


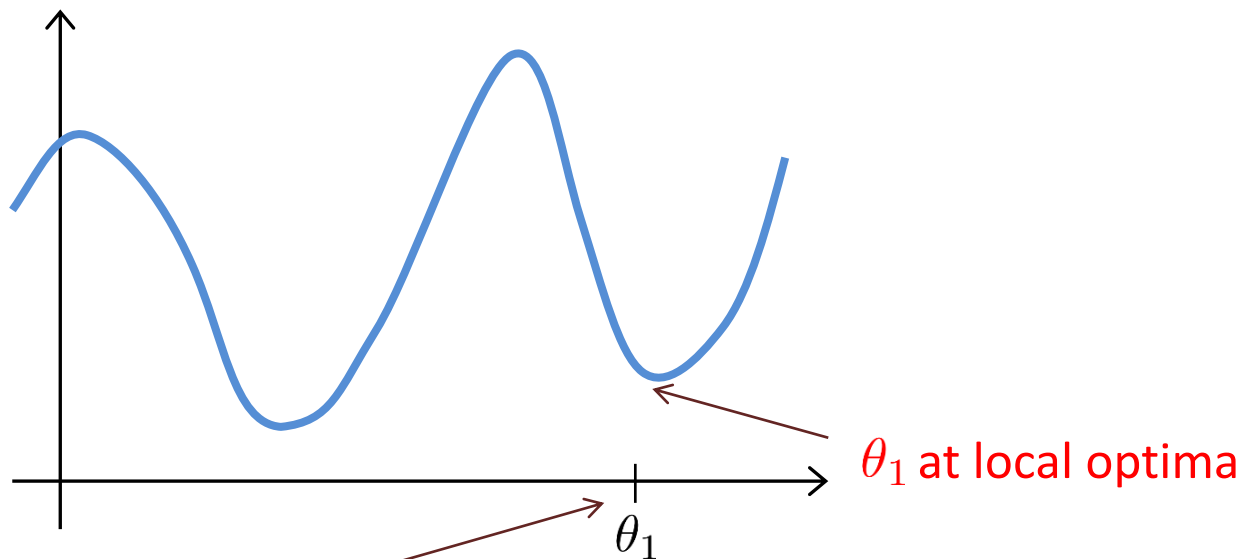
$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_1)$$

If α is too small, gradient descent can be slow.

If α is too large, gradient descent can overshoot the minimum. It may fail to converge, or even diverge.







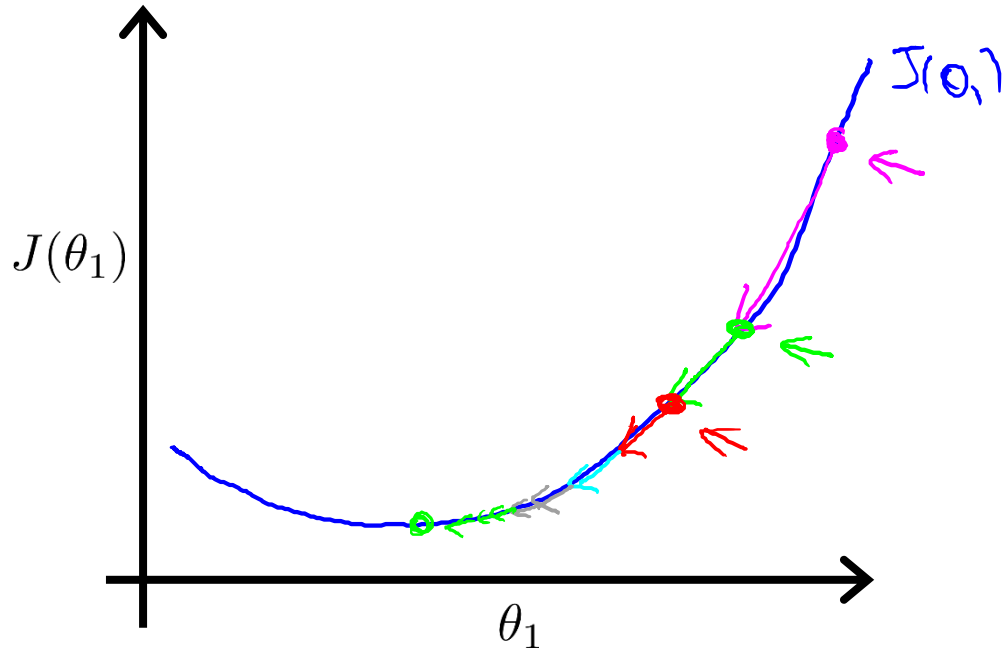
Current value of θ_1

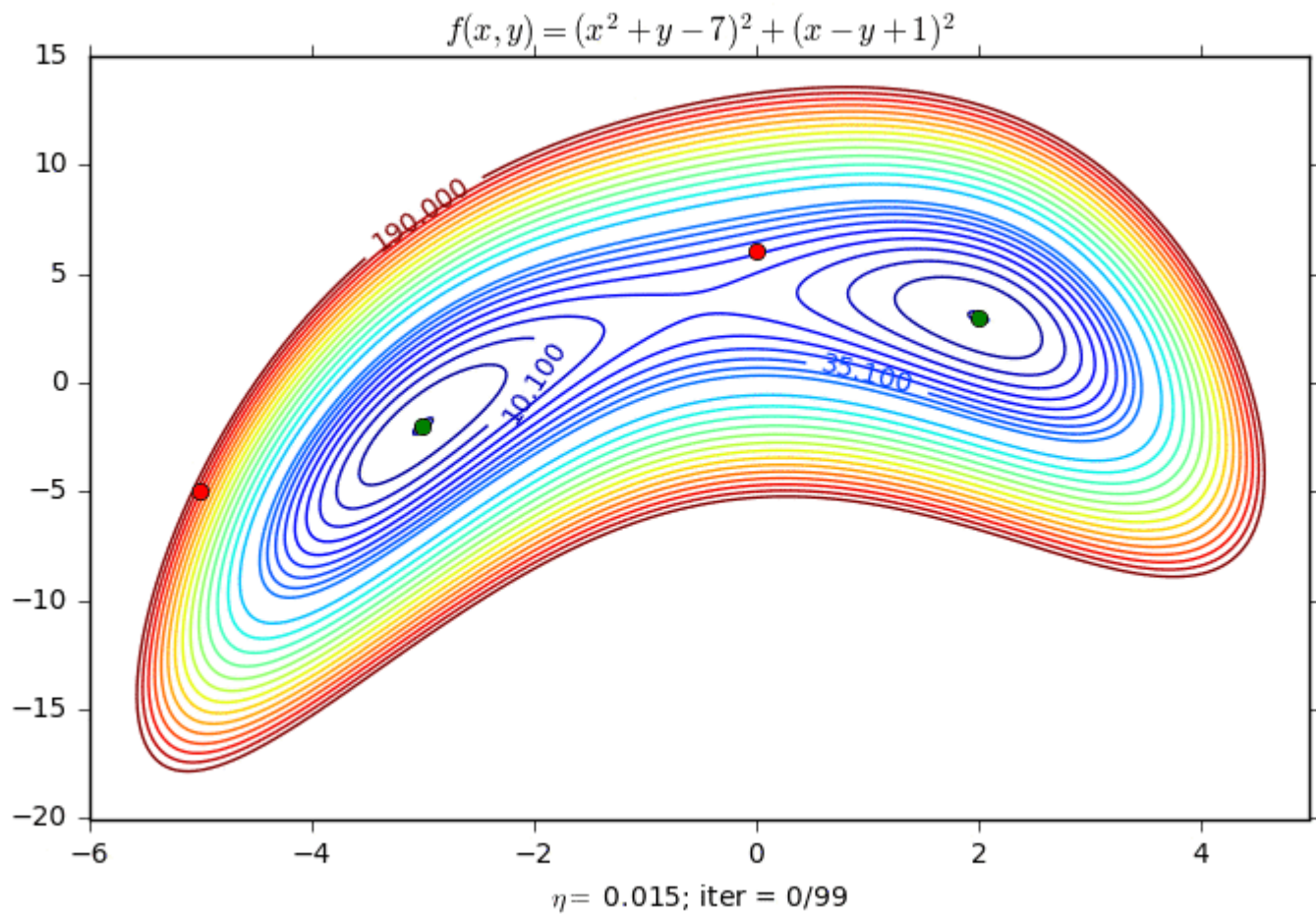
$$\theta_1 := \theta_1 - \alpha \frac{d}{d\theta_1} J(\theta_1)$$

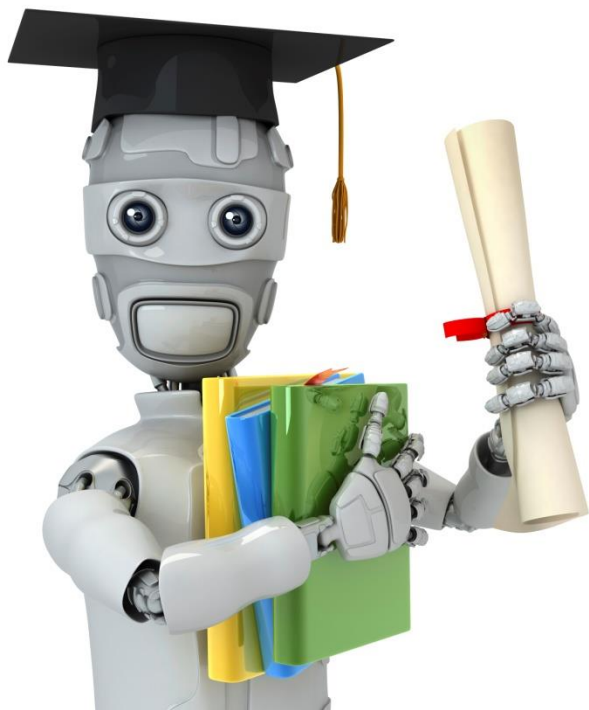
Gradient descent can converge to a local minimum, even with the learning rate α fixed.

$$\theta_1 := \theta_1 - \alpha \frac{d}{d\theta_1} J(\theta_1)$$

As we approach a local minimum, gradient descent will automatically take smaller steps. So, no need to decrease α over time.







Machine Learning

Linear regression with one variable

Gradient descent for linear regression

Gradient descent algorithm

repeat until convergence {
 $\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$
 (for $j = 1$ and $j = 0$)
}

Linear Regression Model

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$\begin{aligned}\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) &= \frac{d}{d\theta_j} \cdot \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 \\ &= \frac{d}{d\theta_j} \cdot \frac{1}{2m} \sum_{i=1}^m (\theta_0 + \theta_1 \cdot x^{(i)} - y^{(i)})^2\end{aligned}$$

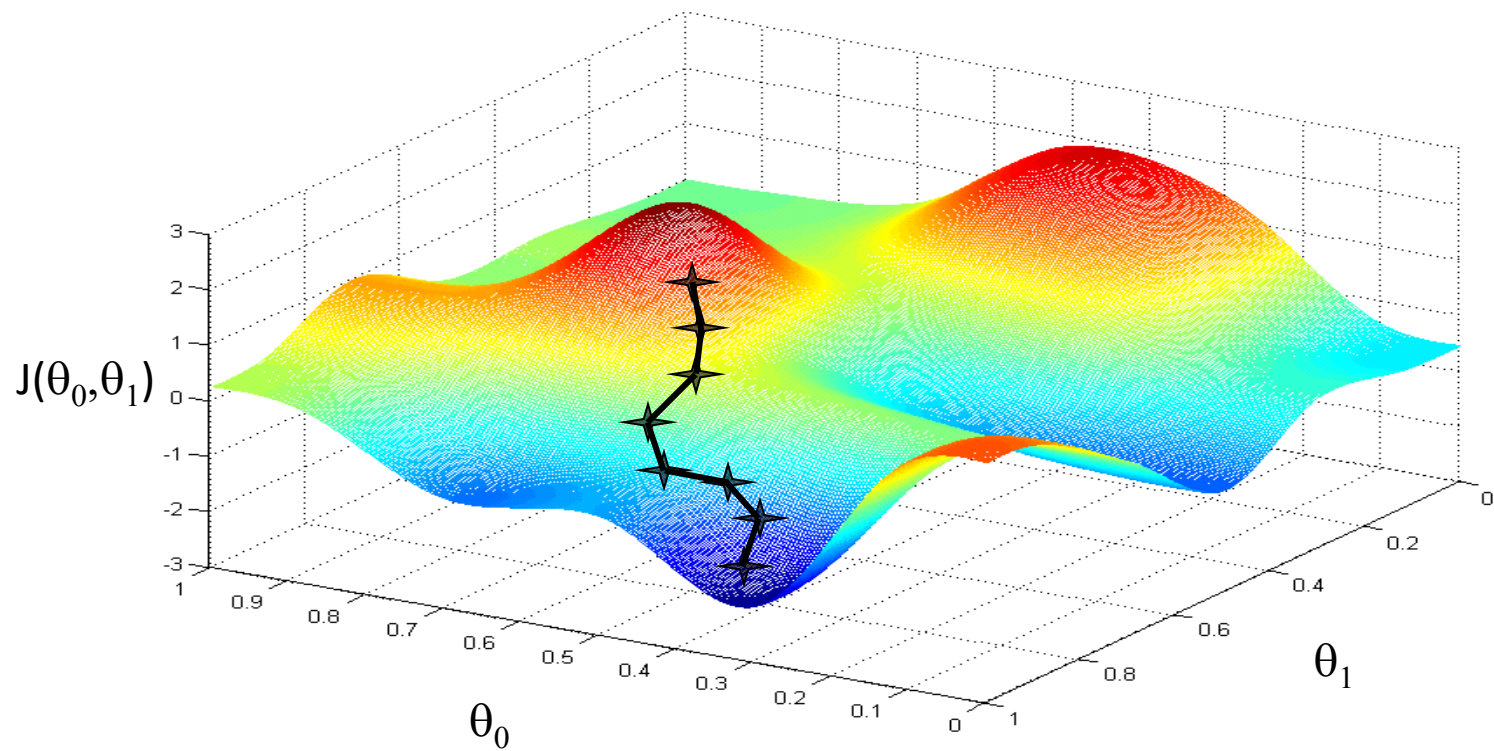
$$j = 0 : \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

$$j = 1 : \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

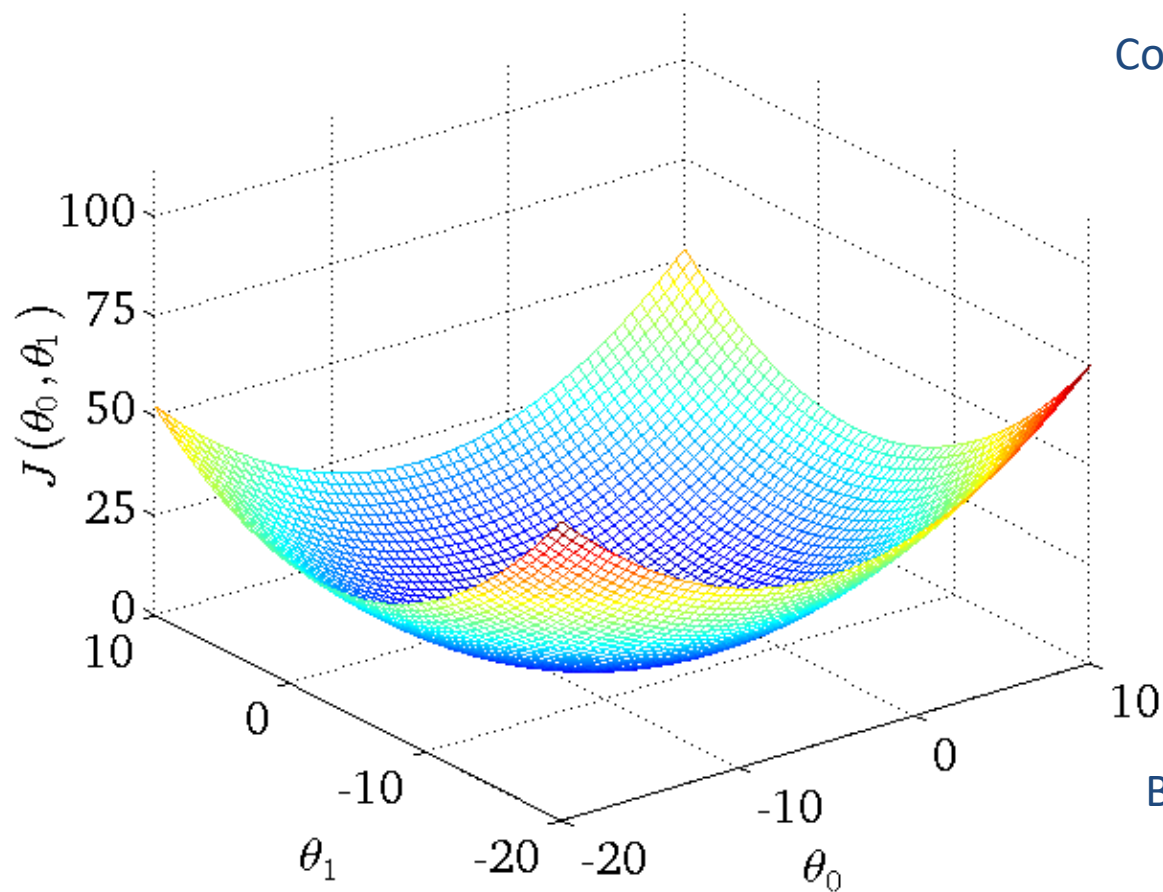
Gradient descent algorithm

$$\begin{array}{l} \text{repeat until convergence } \left\{ \begin{array}{l} \theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \\ \theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)} \end{array} \right\} \end{array}$$

$\frac{d}{d\theta_0} \cdot J(\theta_0, \theta_1)$
update θ_0 and θ_1 simultaneously
 $\frac{d}{d\theta_1} \cdot J(\theta_0, \theta_1)$



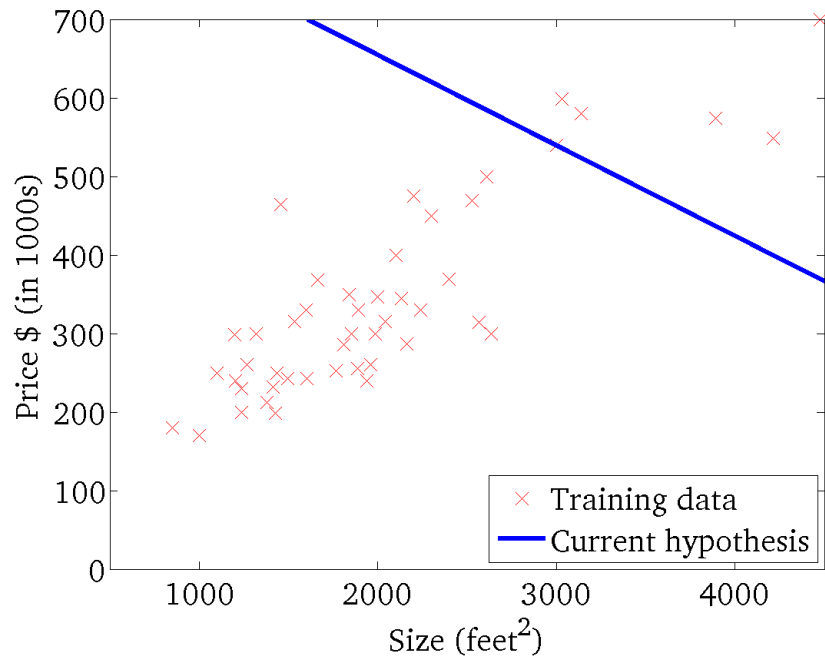
Convex function



Bowl-shaped

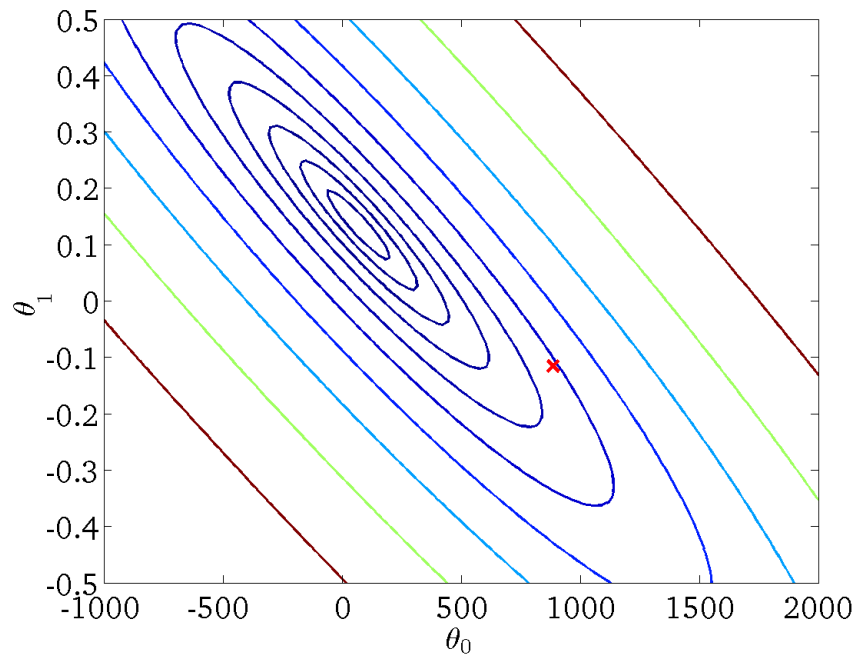
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



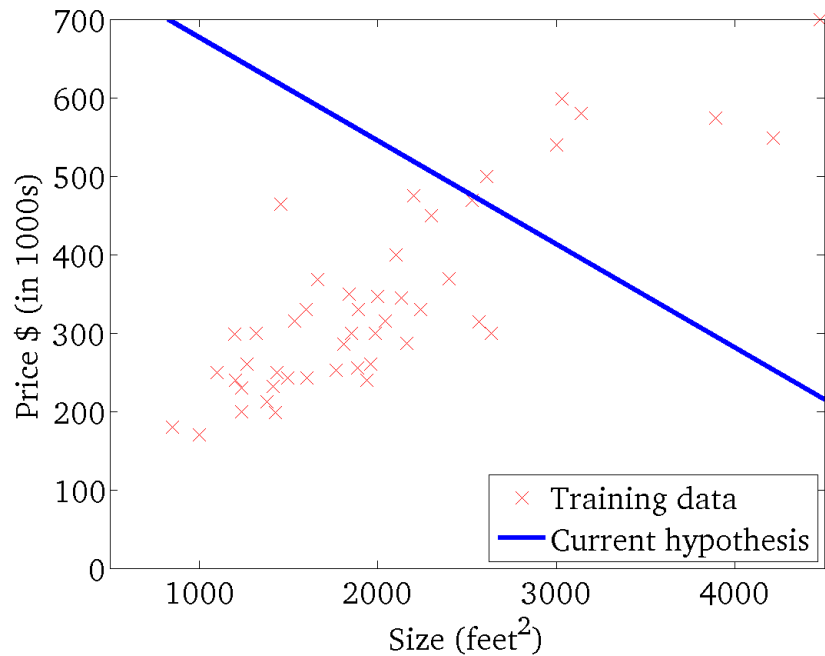
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



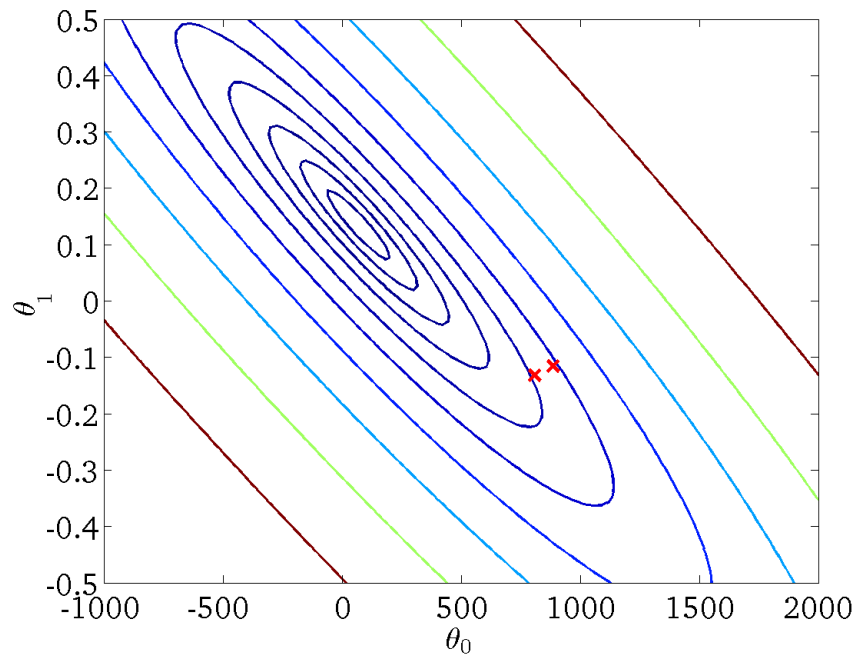
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



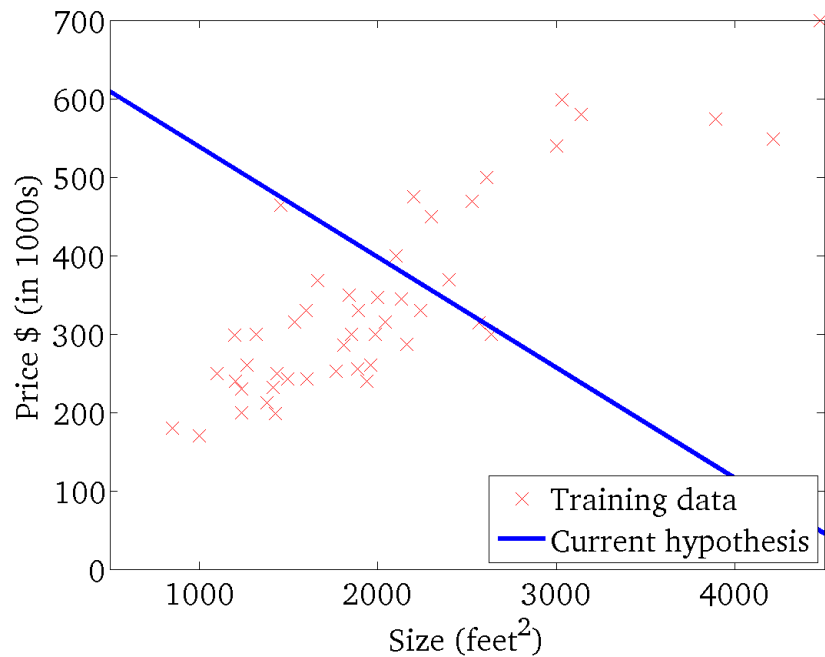
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



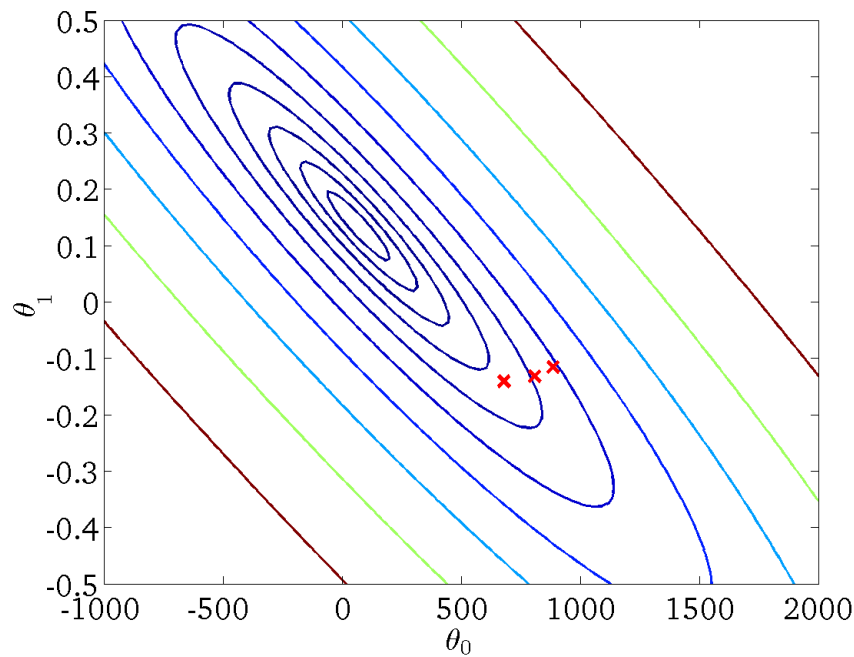
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



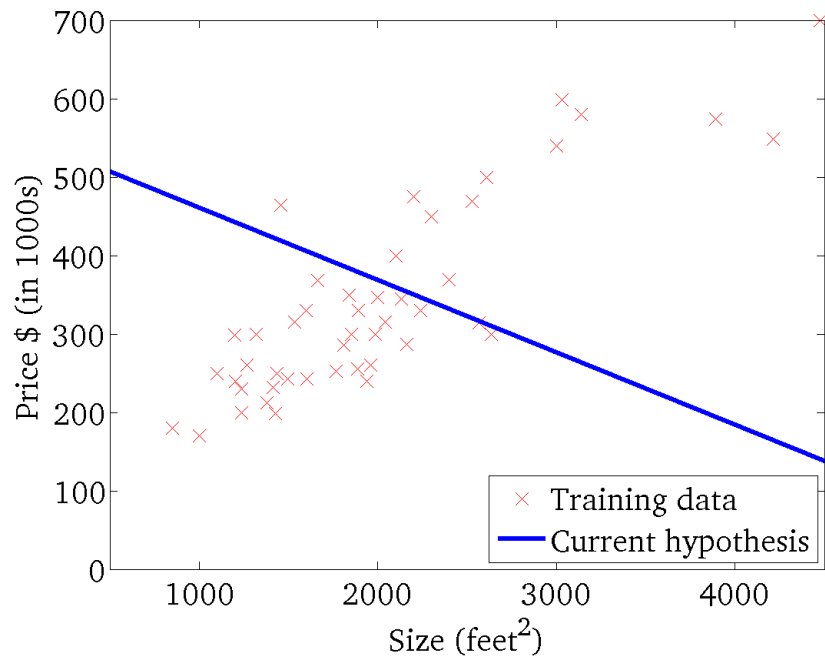
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



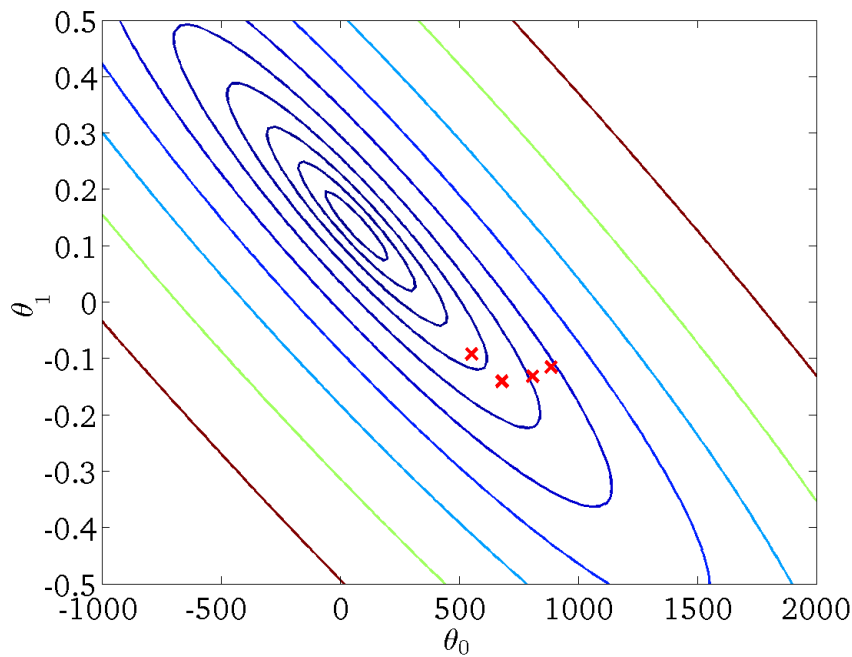
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



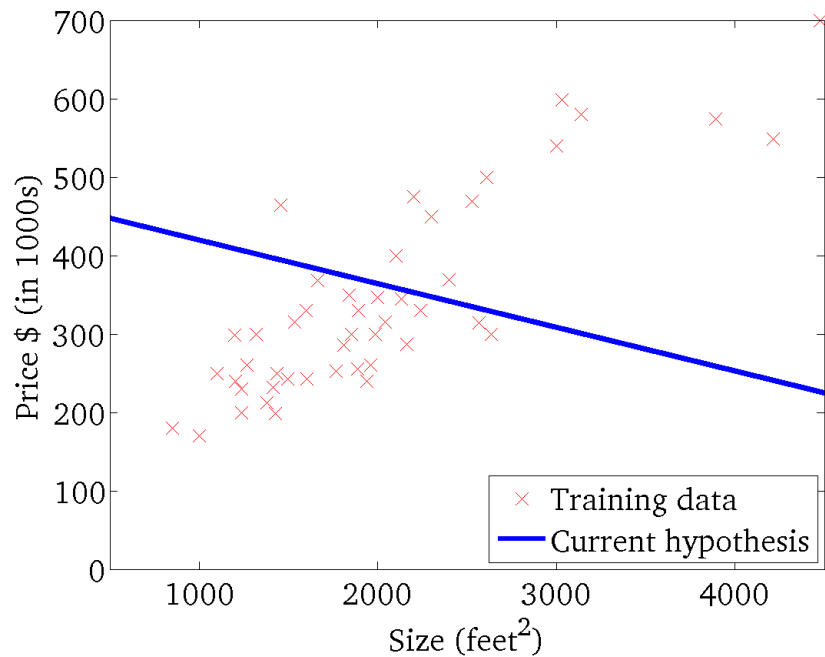
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



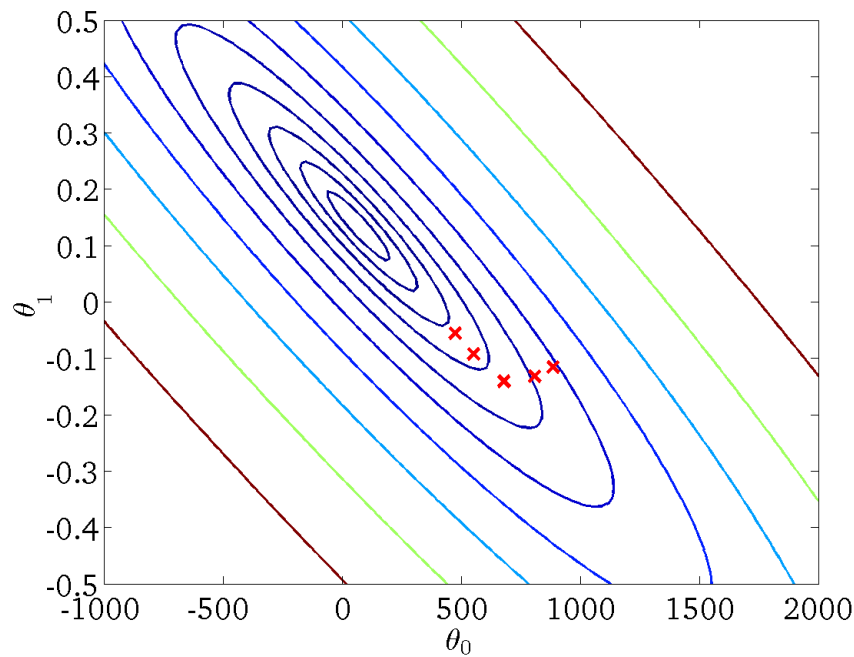
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



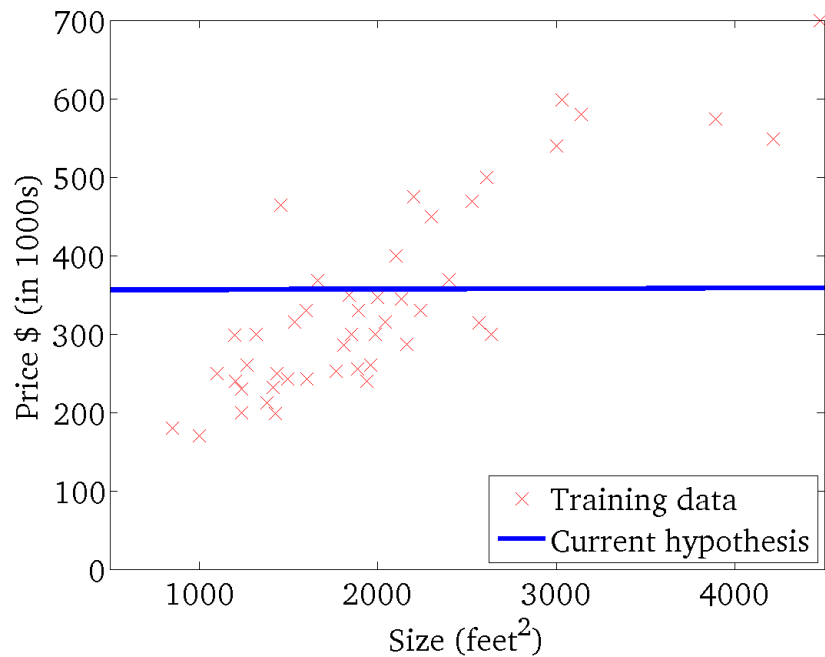
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



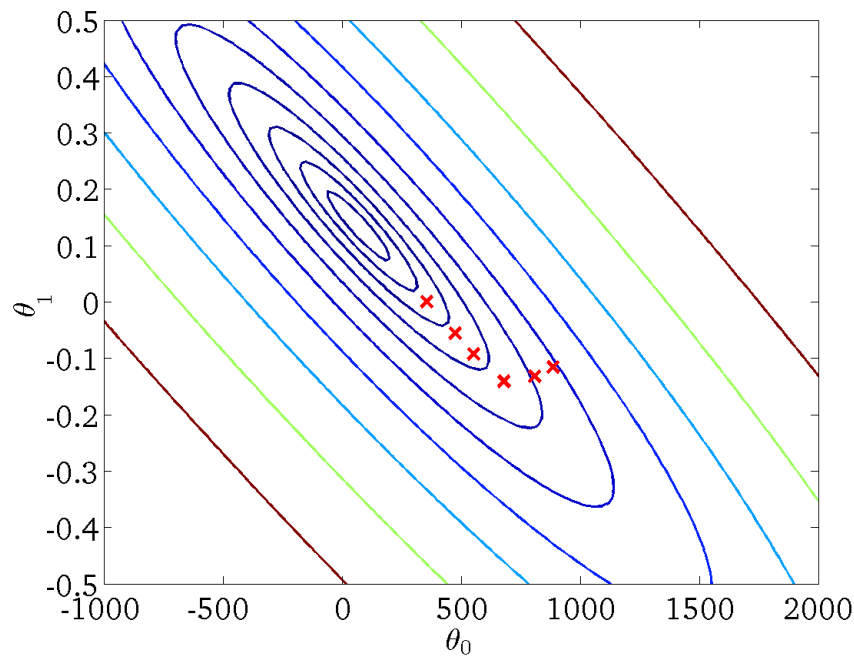
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



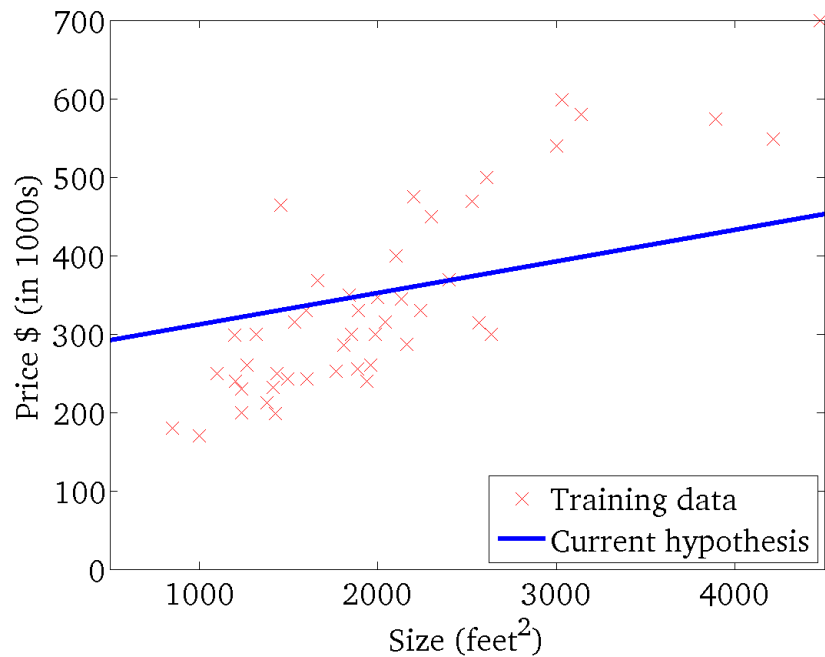
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



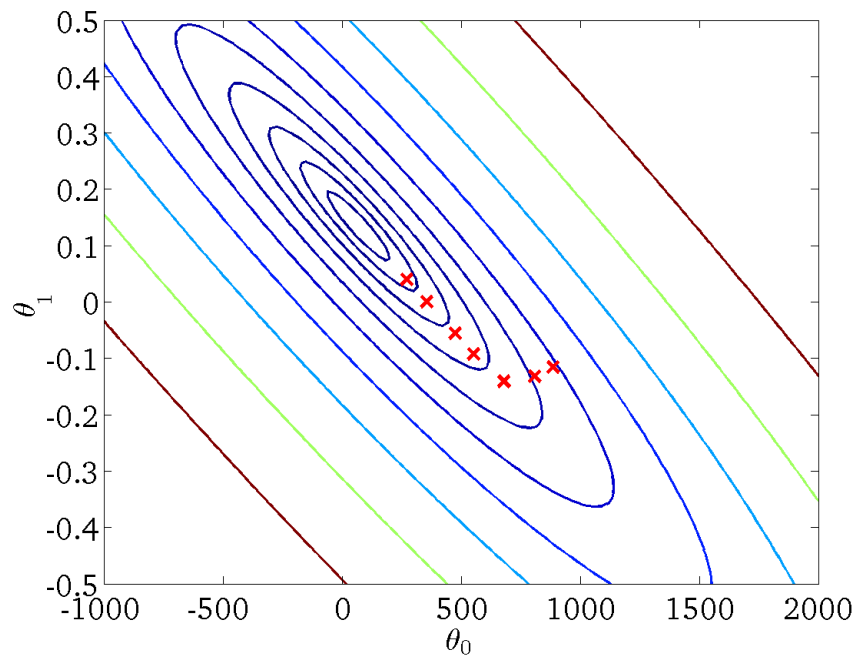
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



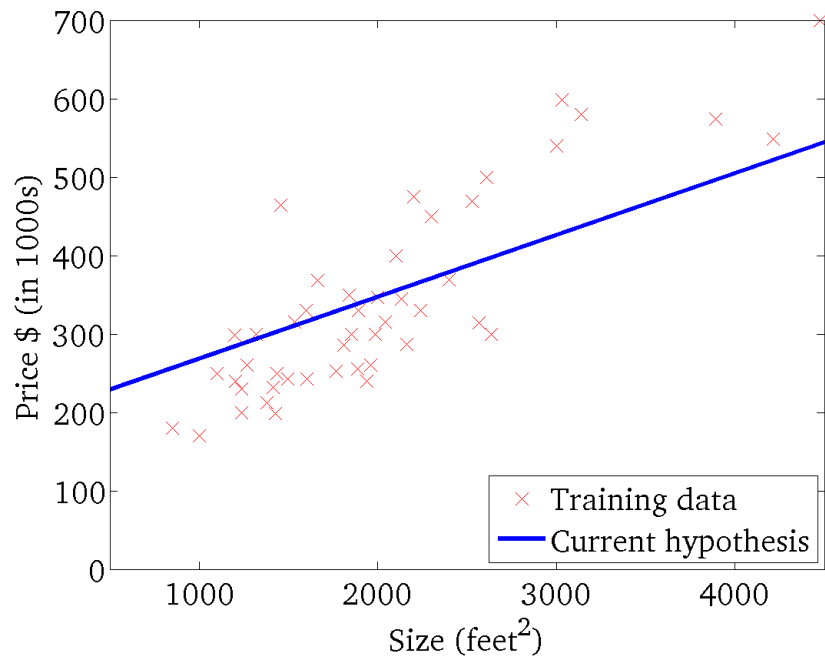
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



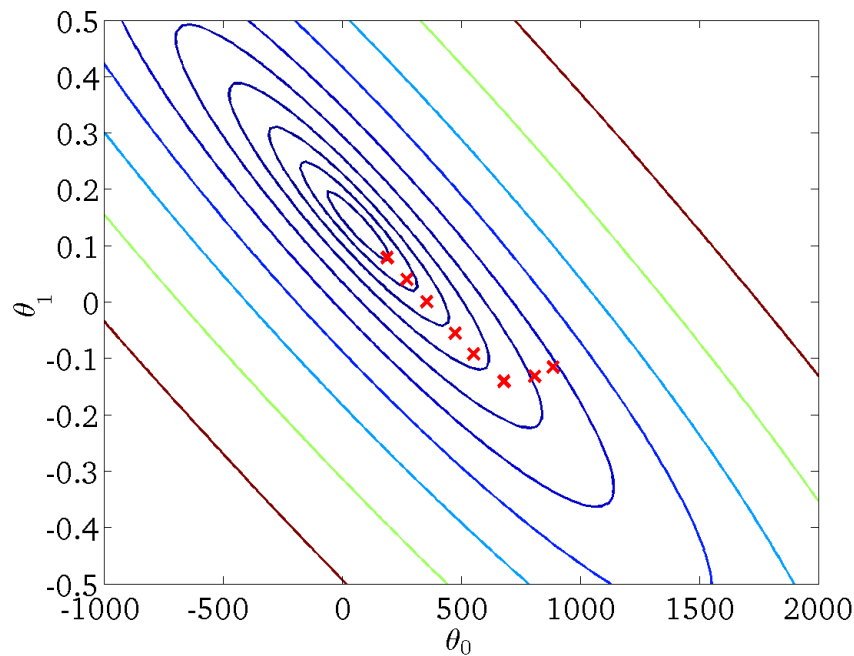
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



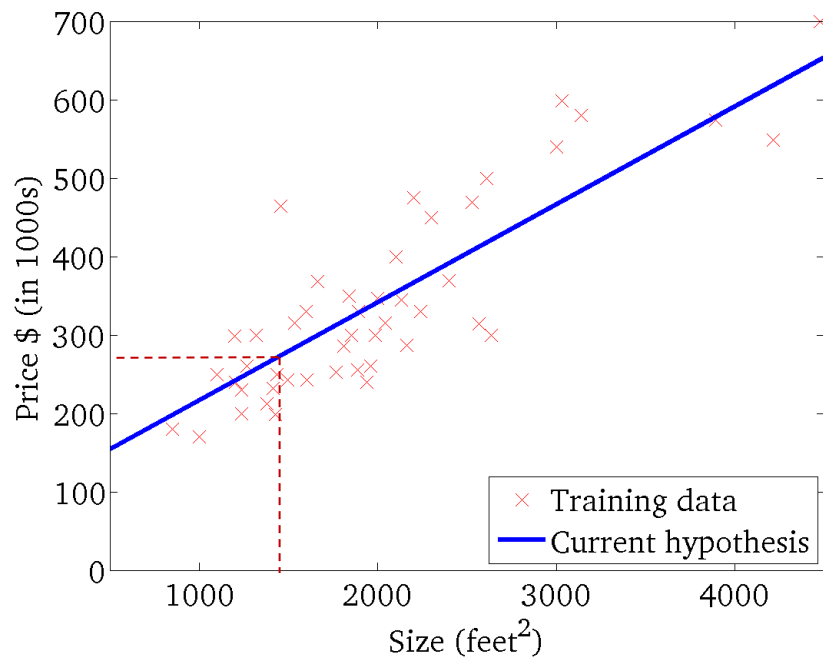
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



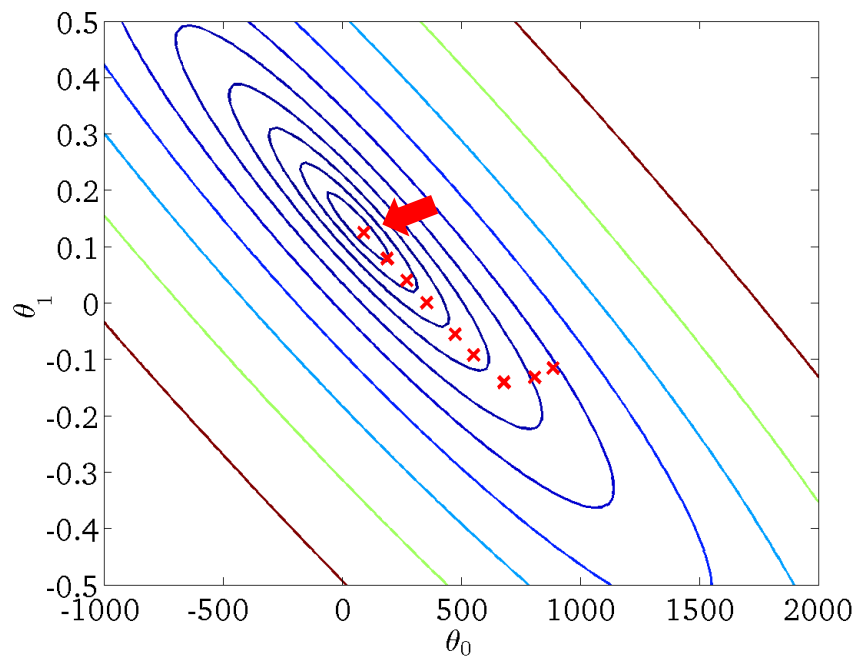
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



“Batch” Gradient Descent

“Batch”: Each step of gradient descent uses all the training examples.